



Energy Conversion I

alexis.martin@ensea.fr
Desk D216





Layout

- Non isolated choppers
- Switch mode power supplies
- **Power components**
 - Passive components
 - Inductor
 - Transformer
 - Capacitor
 - Active components
 - Diodes
 - Transistors
 - Protections
 - Fuses
 - Thermistors
 - ESD protection

**Passive
components**

**Active
components**

Protections

Passive components

Passive components

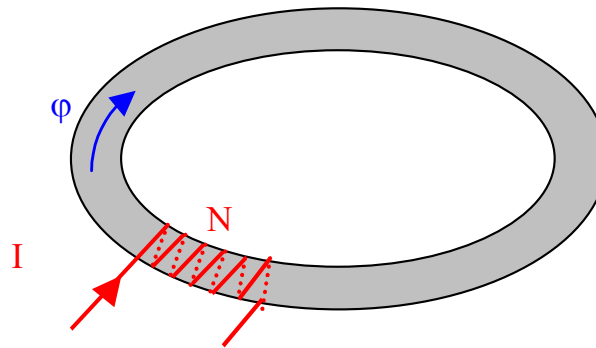
Inductor

Transformer
Capacitor

Active components

Protections

Iron core Inductor



$\phi = B.S$: elementary flux (real flux)

$\Phi = N.\phi = L.I$: “total flux” (sawn by the electrical circuit)

$$\int_{\text{circuit}} \mathbf{H} \cdot d\mathbf{l} = N.I$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu}$$

$$N.I = \left[\int_{\text{circuit}} \frac{dl}{\mu.S} \right] \cdot \phi = R. \phi$$

$$L_1 = \frac{N^2}{R} = N^2 \cdot A_L$$

$$A_L = \frac{1}{\int_{\text{circuit}} \frac{dl}{\mu S}} \approx \frac{\mu S}{l_{\text{circuit}}}$$

Passive components

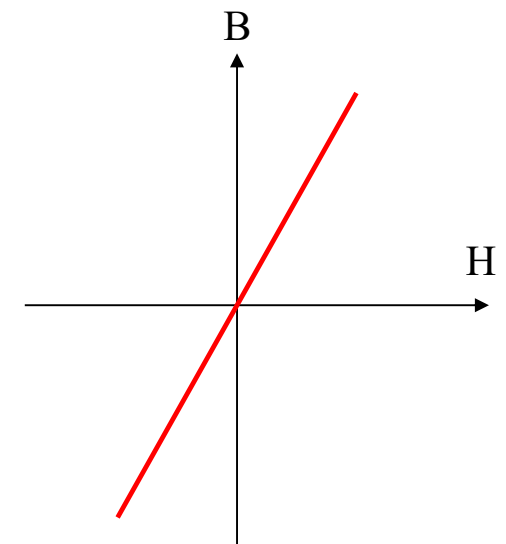
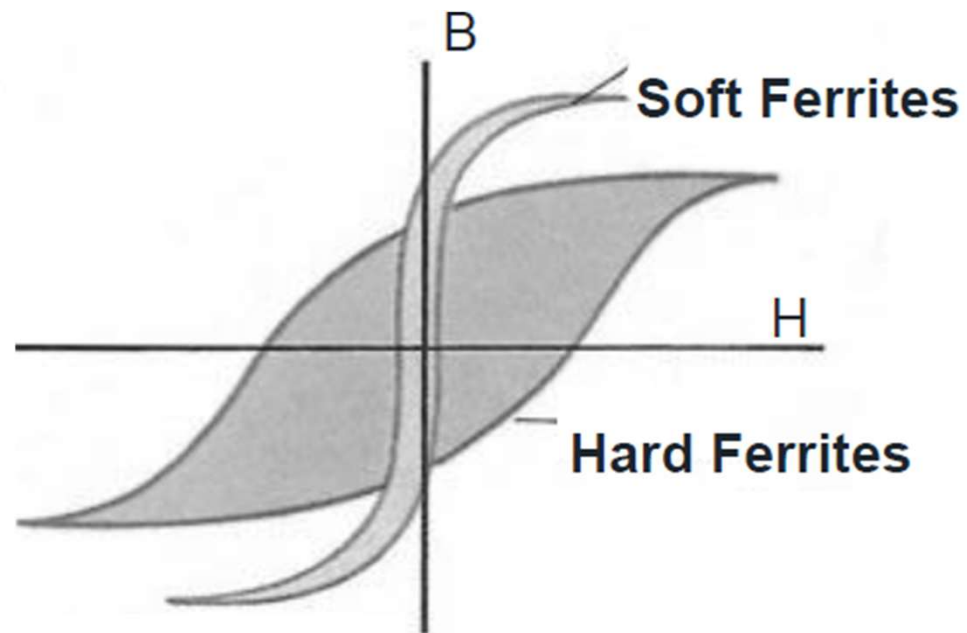
Inductor

Transformer
Capacitor

Active components

Protections

Hysteresis



*Ideal
Characteristic*

$$\mu_r = \frac{1}{\mu_0} \frac{\partial B}{\partial H}$$

EMC BASICS, Würth Elektronik

Passive components

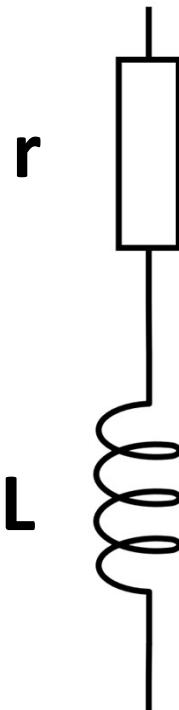
Inductor

Transformer
Capacitor

Active components

Protections

Losses



Value of relative permeability (and so, inductance) depends on :

- Core material
- Temperature
- Current
- Frequency
- Pressure

$R = R_c + R_I$: Copper losses (wires) + iron losses (magnetic material)

Passive components

Inductor

Transformer
Capacitor

Active components

Protections

Iron core Inductor

- Core material

Typical value ranges

- Iron powder (Fe): $\mu_r = 50 \dots 150$
- Nickel-zinc ferrite ($\text{Ni}_x\text{Zn}_{1-x}\text{Fe}_2\text{O}_4$): $\mu_r = 50 \dots 1500$
- Manganese-zinc ferrite ($\text{Mn}_x\text{Zn}_{1-x}\text{Fe}_2\text{O}_4$): $\mu_r = 2000 \dots 10000$
- Nanocrystalline material (Fenc): $\mu_r = 5000 \dots 95000$

Passive
components

Inductor

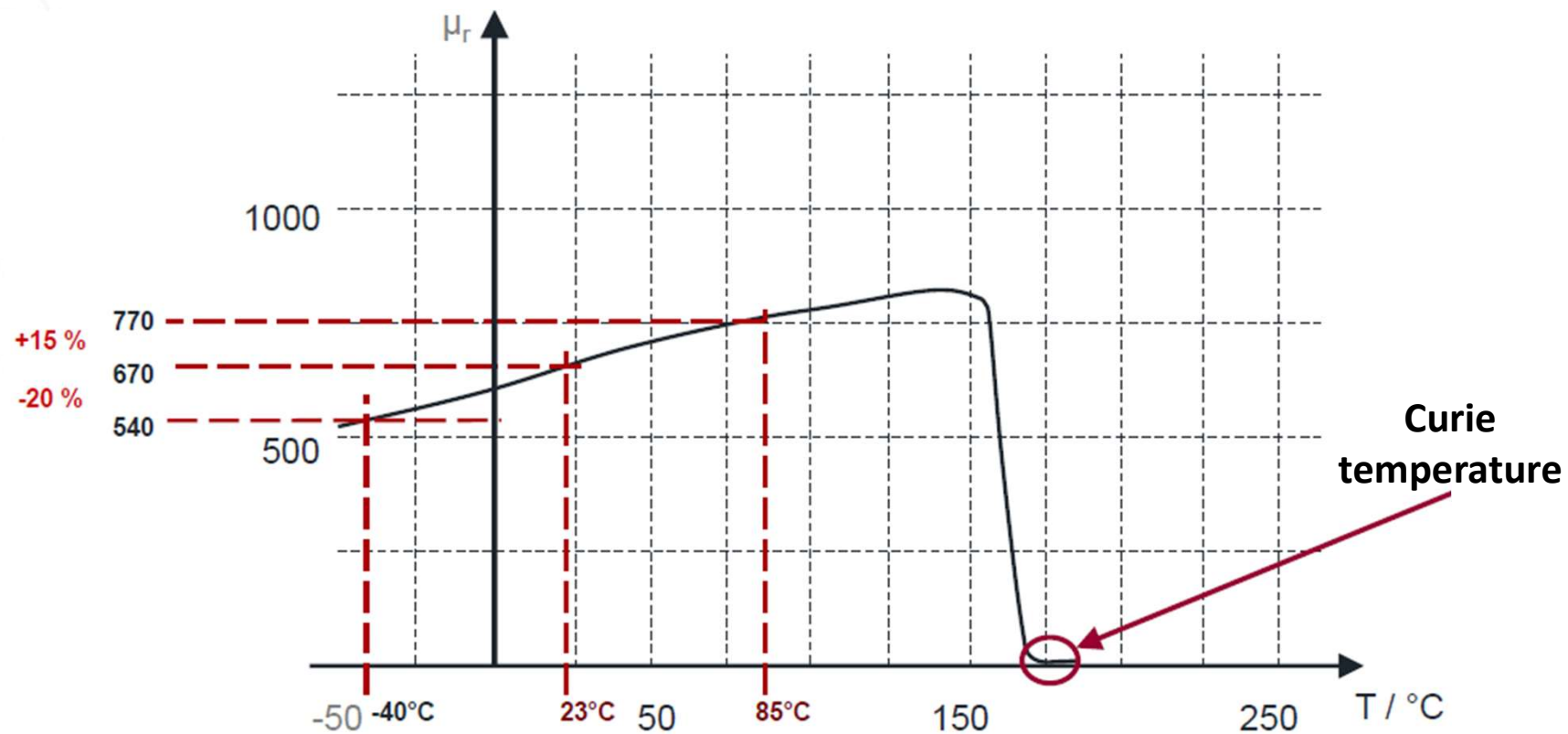
Transformer
Capacitor

Active
components

Protections

Iron core Inductor

- Temperature



EMC BASICS, Würth Elektronik

Passive
components

Inductor

Transformer
Capacitor

Active
components

Protections

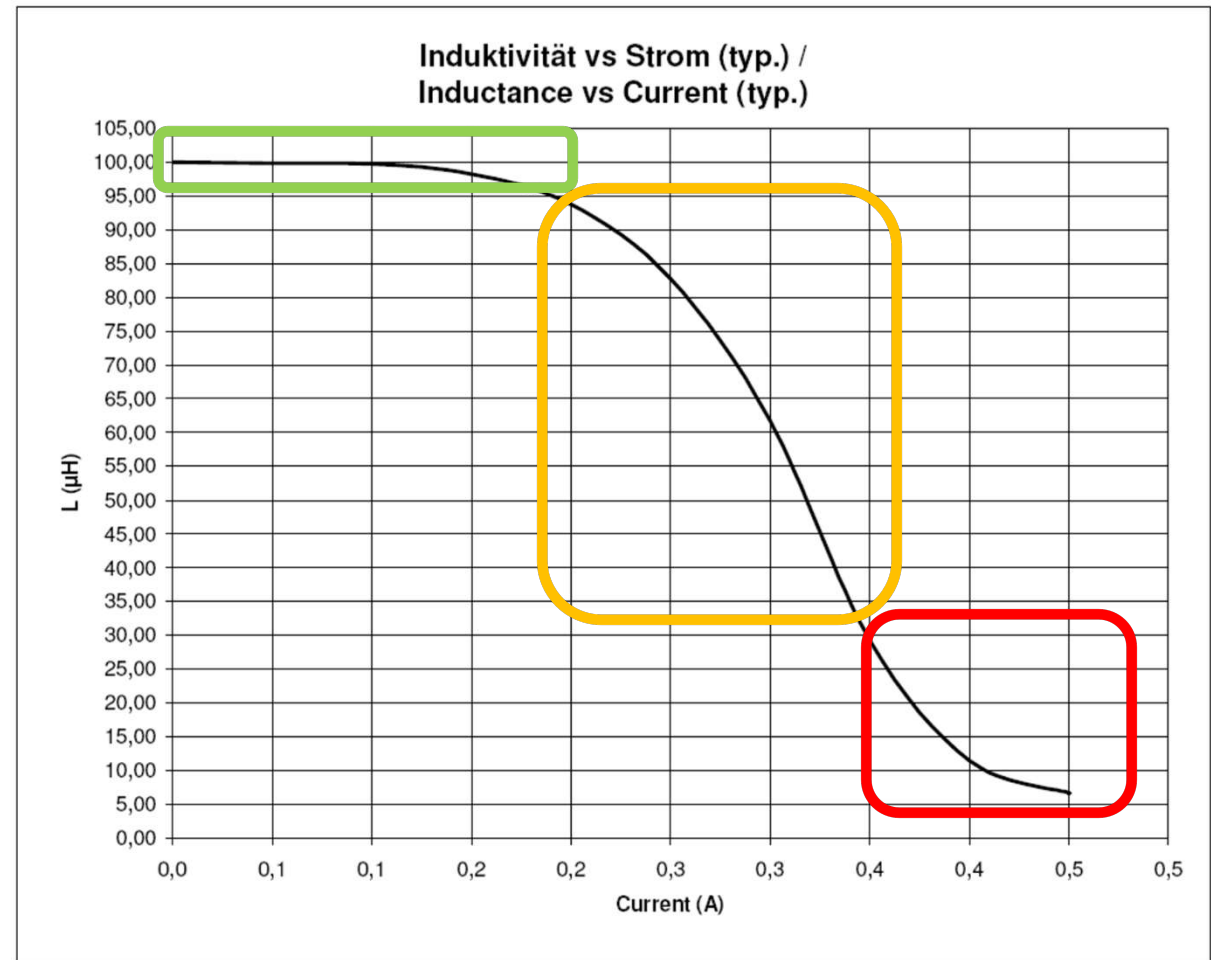
Iron core Inductor

- Current

Safe Operation
Area

Critical
Operating Area

Total Saturation
Area



POWER INDUCTOR SELECTION , Würth Elektronik

Passive
components

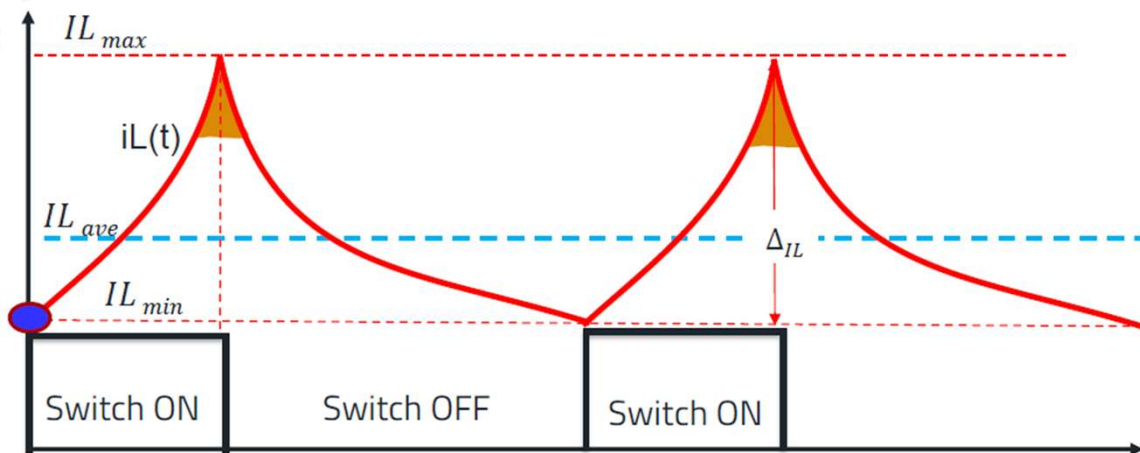
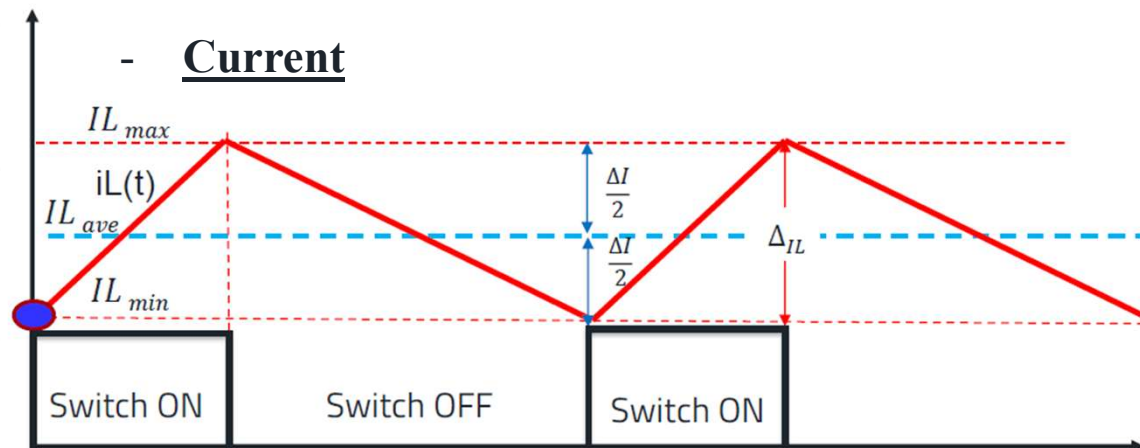
Inductor

Transformer
Capacitor

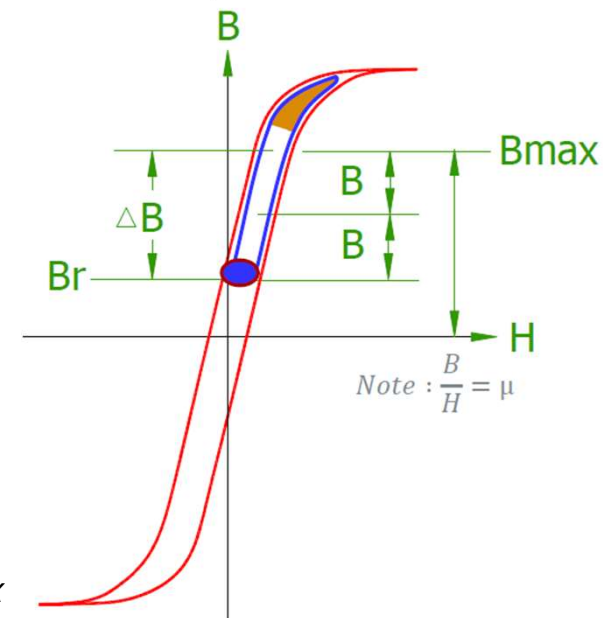
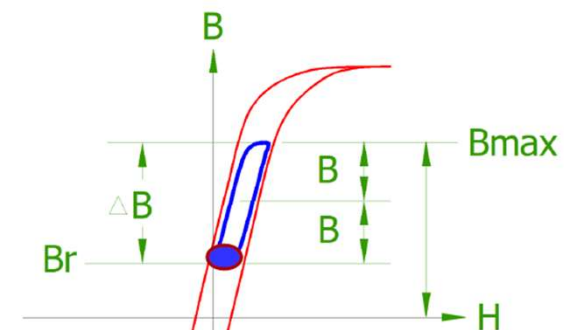
Active
components

Protections

Iron core Inductor



POWER INDUCTOR SELECTION, Würth Elektronik



Passive
components

Inductor

Transformer
Capacitor

Active
components

Protections

Iron core Inductor

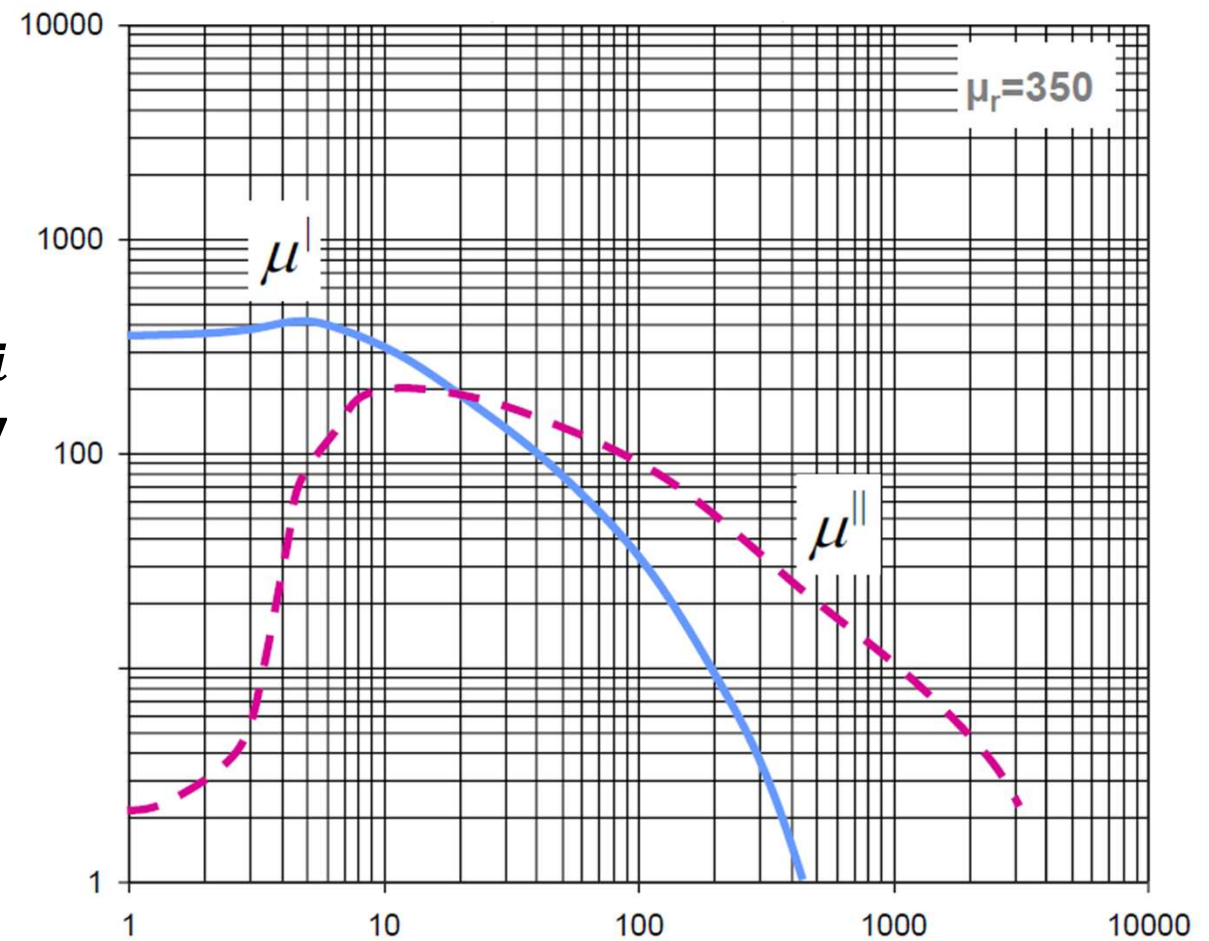
- Frequency



$$R = R_c + R_i$$

$$R_i = \omega L_o \mu''$$

$$L = L_o \mu'$$



EMC BASICS, Würth Elektronik

Passive
components

Inductor

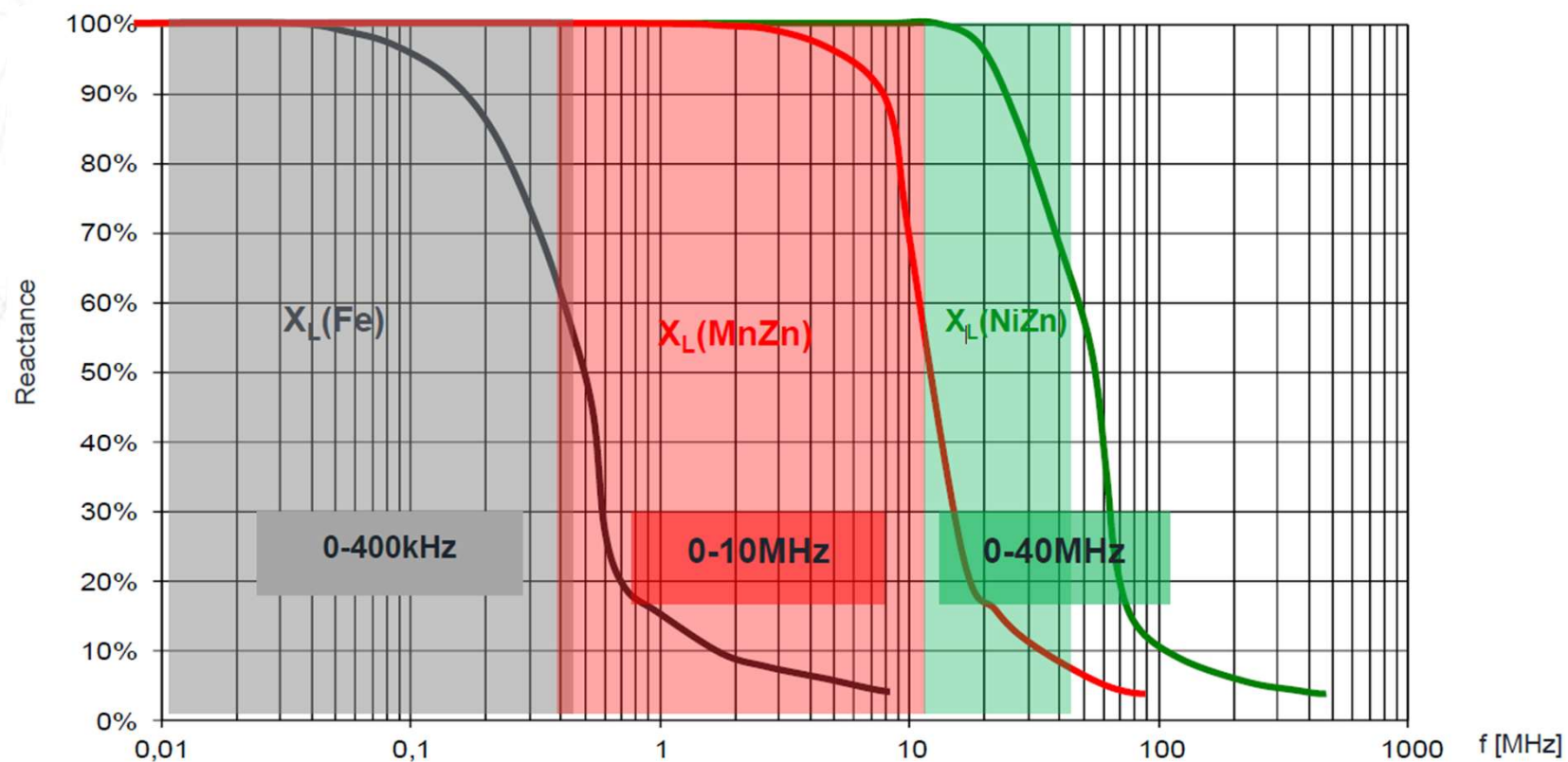
Transformer
Capacitor

Active
components

Protections

Iron core Inductor

- Frequency



EMC BASICS, Würth Elektronik

Passive components

Inductor

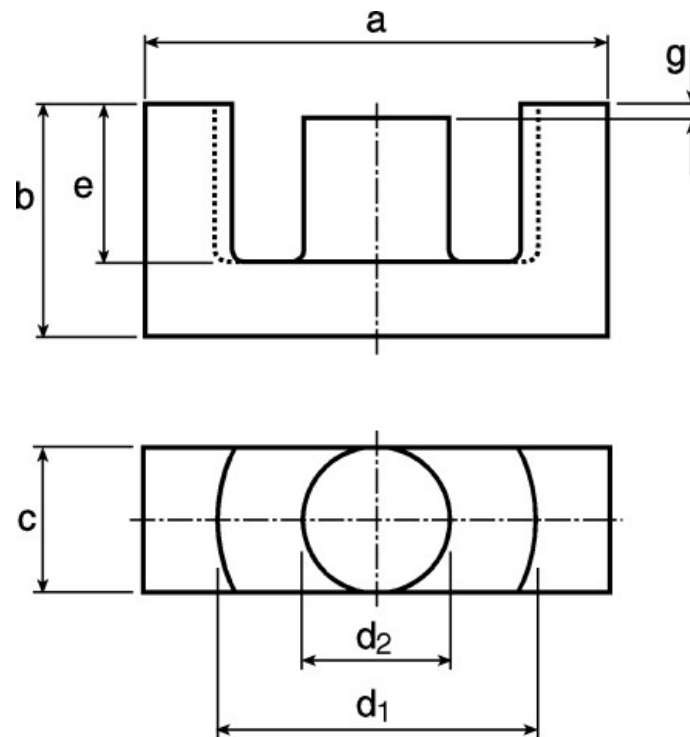
Transformer

Capacitor

Active components

Protections

Magnetic circuit example



ETD series



Passive components

Inductor

Transformer
Capacitor

Active components

Protections

Magnetic circuit example



- Toroidal coil

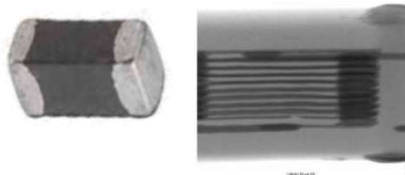


- Solenoid coil

- Solenoid coil + Shielding



- Multilayer (low current)



Passive components

Inductor

Transformer

Capacitor

Active components

Protections

Stored energy – Air-gap

Magnetic energy density stored

$$\omega_{mag} = \frac{B^2}{2 \cdot \mu_0 \cdot \mu_r}$$

Example : Ferrinox B52

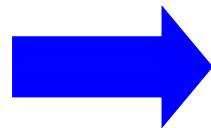
$B = 0,4 \text{ T}$

$\mu = \mu_0 \cdot \mu_r = 2,5 \cdot 10^{-3} \text{ H.m}^{-1}$

$$\omega_{mag} = 32 \text{ J/m}^3$$

Air-gap : $\mu_0 = 4\pi \cdot 10^{-7} \text{ H.m}^{-1}$

$$\omega_{mag} = 65 \text{ kJ/m}^3$$



Store energy in the air-gap

Passive components

Inductor

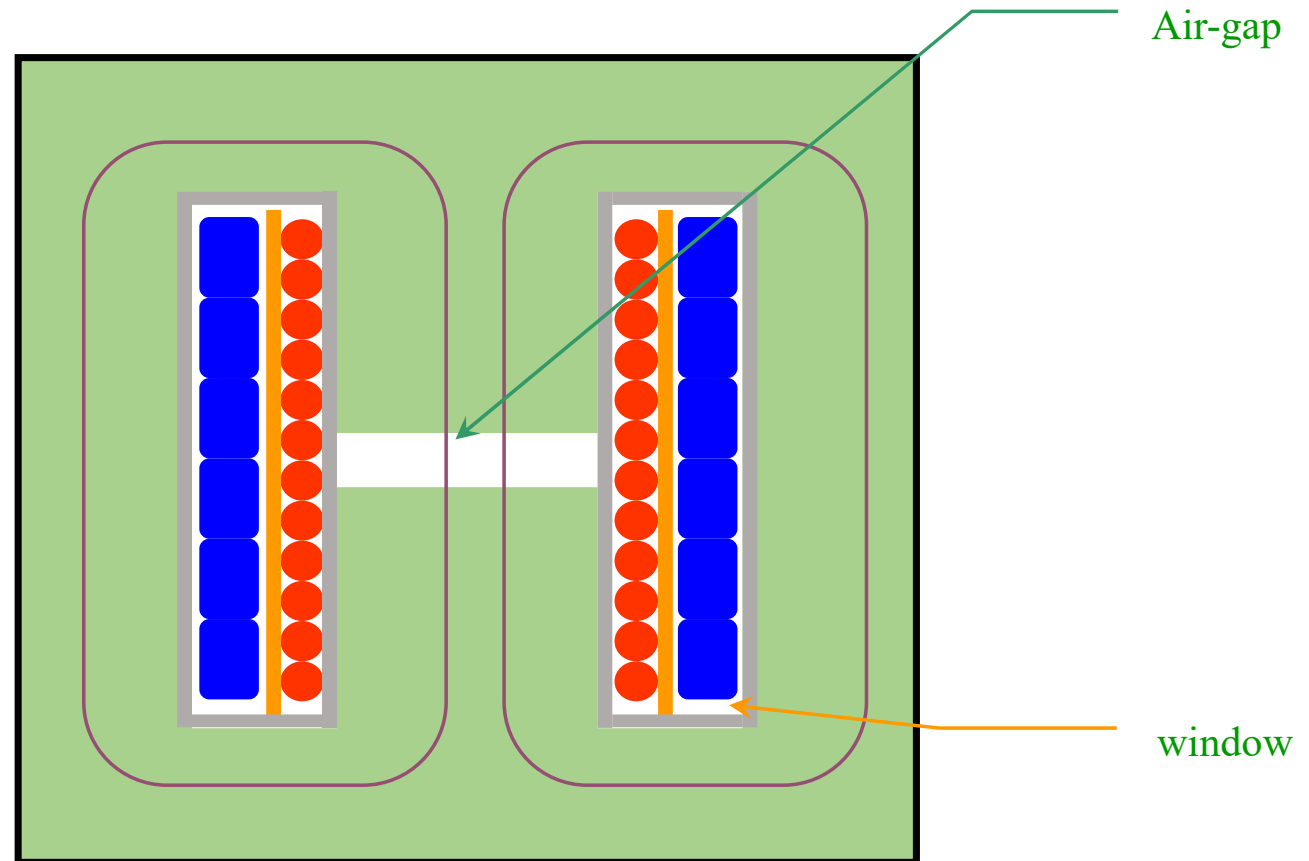
Transformer

Capacitor

Active components

Protections

With air-gap



Passive components

Inductor

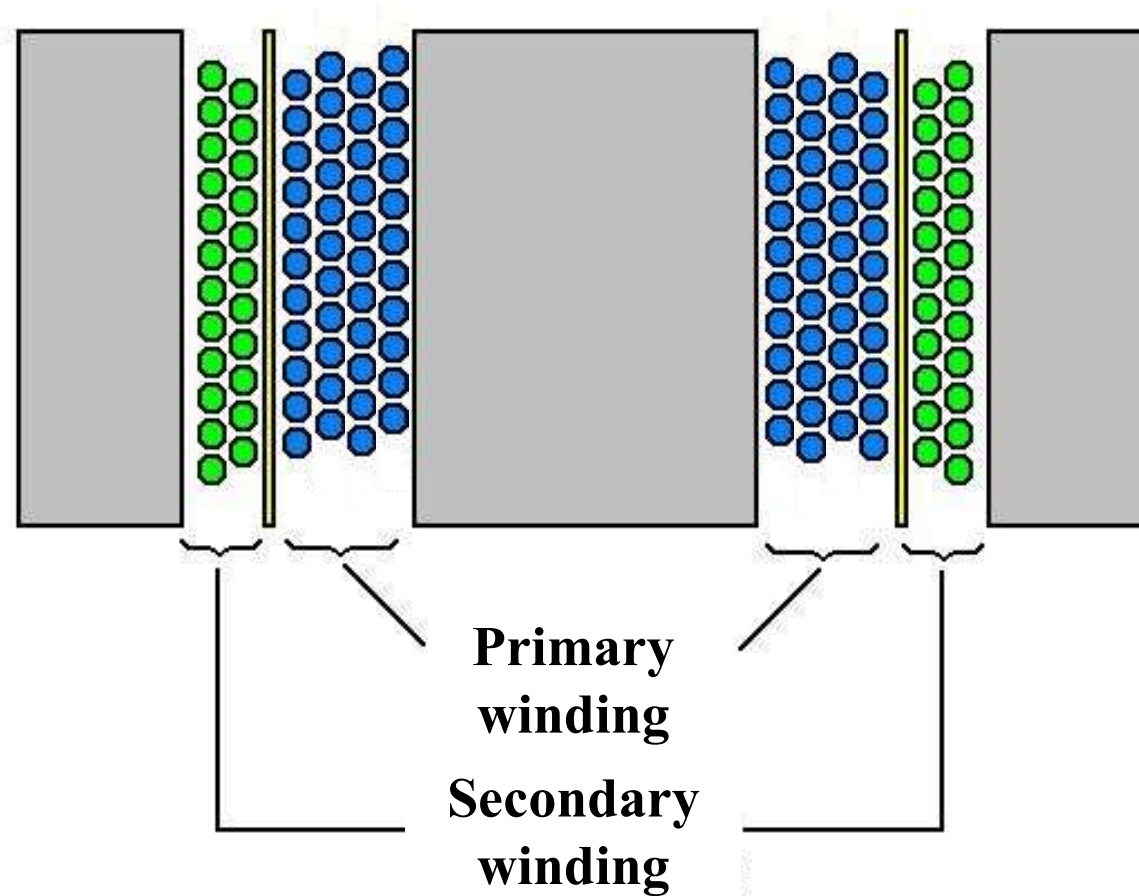
Transformer

Capacitor

Active components

Protections

Closed core



Passive components

Inductor

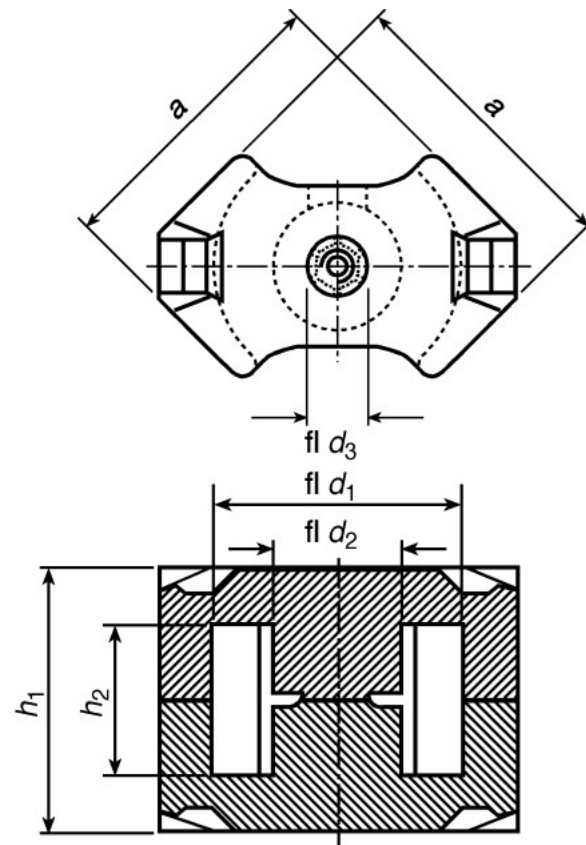
Transformer

Capacitor

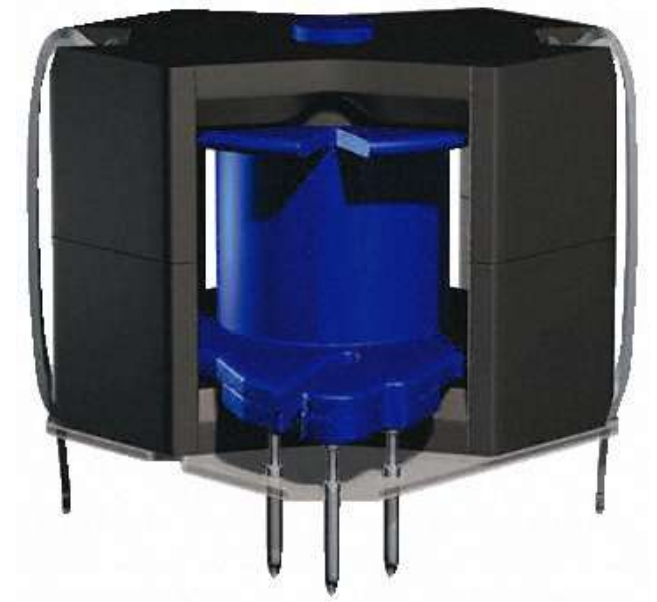
Active components

Protections

Magnetic pot core



RM- 8- bis RM-14-Kernstze
fr nichtlineare Drosselspulen



RM series

Passive components

Inductor

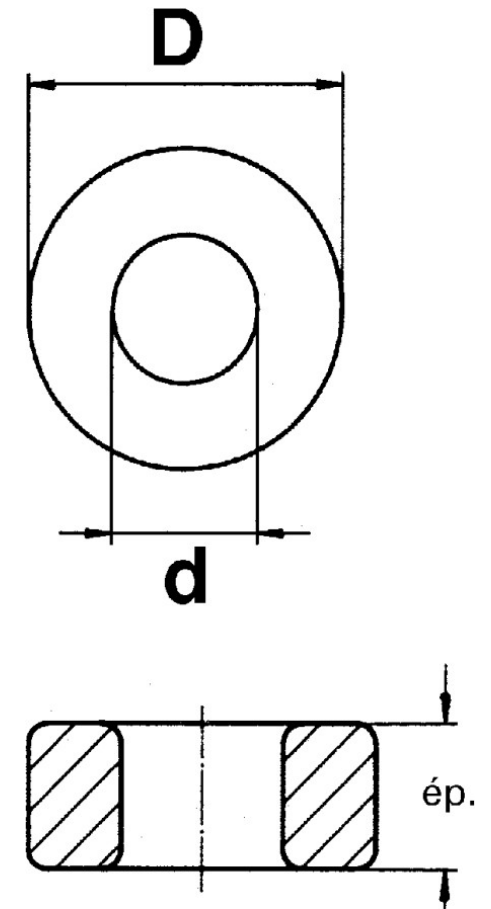
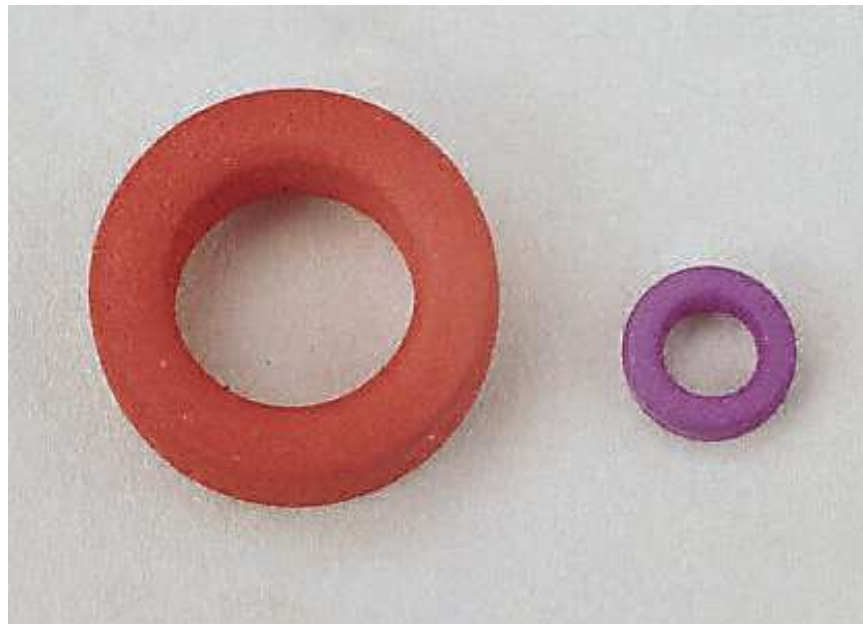
Transformer

Capacitor

Active components

Protections

Torus



Passive components

Inductor

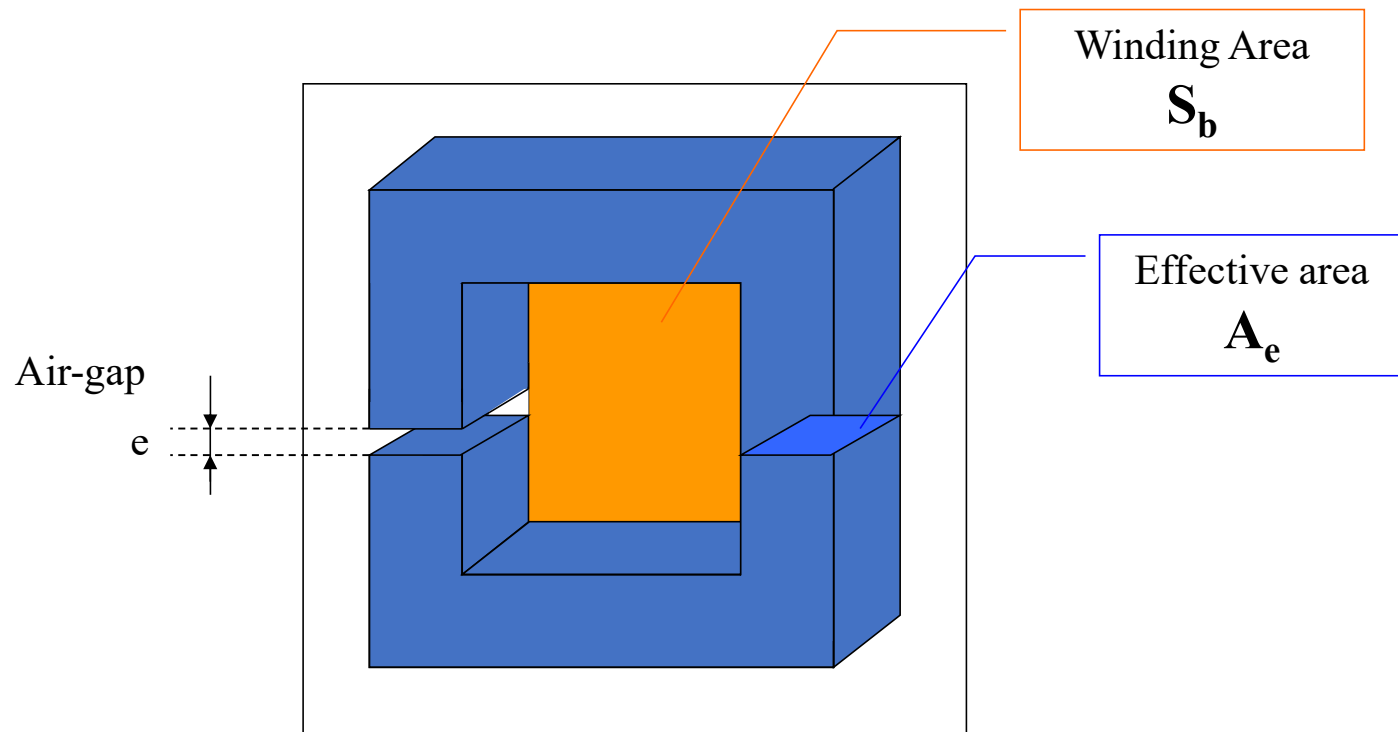
Transformer

Capacitor

Active components

Protections

Magnetic circuits parameters



Inductance factor: $\mathbf{A_L}$ = Inductance obtained with ONE loop

Passive components

Inductor

Transformer

Capacitor

Active components

Protections

Critical constraints

*Induction in the material: $B < B_{sat}$
0,2 to 0,5 T for classical materials*

*Current density in the winding: $\delta < \delta_{max}$
5 A/mm² for copper*

Sizing : product $A_e S_b$

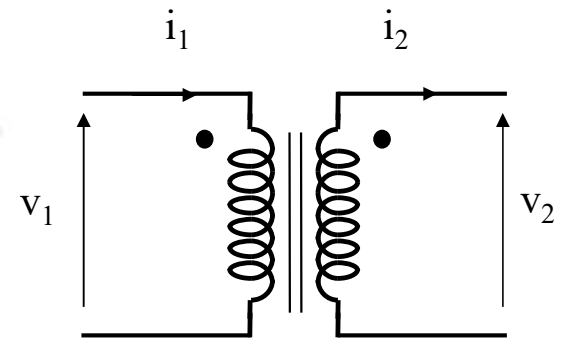
Passive components

Inductor
Transformer
Capacitor

Active components

Protections

Transformer



Primary : N_1 turns
Secondary : N_2 turns

Winding direction

$$v_1 = N_1 \cdot \frac{d\phi}{dt}$$

$$v_2 = N_2 \cdot \frac{d\phi}{dt}$$

$$m = \frac{N_2}{N_1} = \frac{v_2}{v_1}$$

$$\oint H \cdot dl = R \cdot \phi = N_1 \cdot i_1 - N_2 \cdot i_2$$

Magnetizing current i_{10} : Hopkinson's law

$$R \cdot \phi = N_1 \cdot i_{10}$$

$$\Phi = L_1 \cdot i_{10} \quad L_1 = \frac{N_1^2}{R}$$

$$i_1 - m \cdot i_2 = i_{10}$$

$$v_1 = L_1 \cdot \frac{di_{10}}{dt}$$

Passive components

Inductor

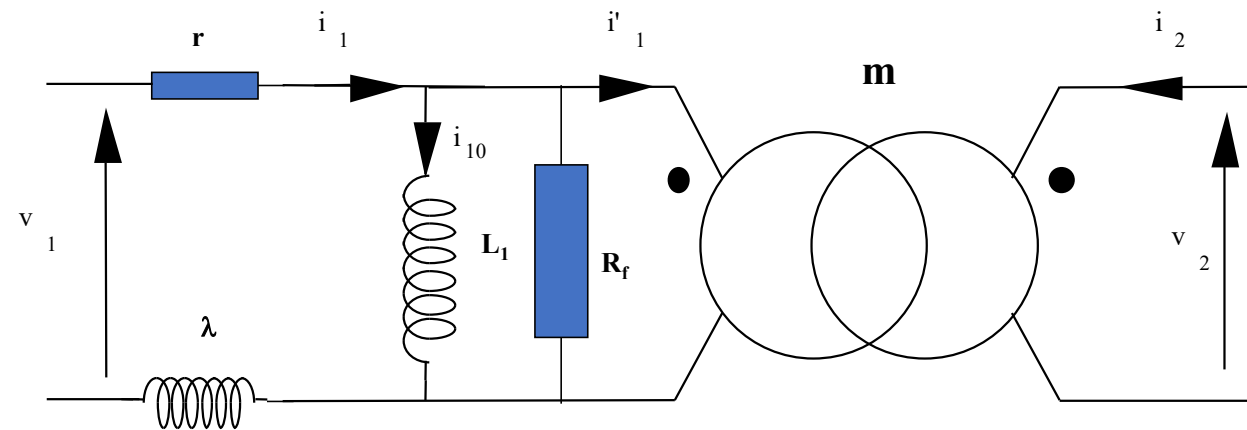
Transformer

Capacitor

Active components

Protections

Transformer : equivalent scheme



L_1 : Magnetizing inductance (at primary side)

R_f : ferromagnetic losses

r : Joule loss (or ohmic loss)

λ : leakage inductance

Passive components

Inductor

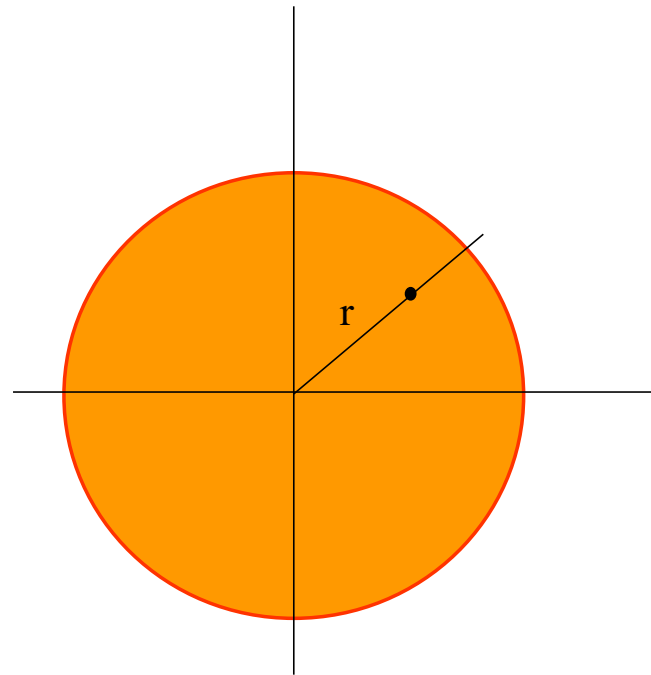
Transformer

Capacitor

Active components

Protections

Skin effect



$$j(r) = j \cdot J_0 \left(e^{i\pi \frac{r}{\delta}} \right).$$

$$e_d = \frac{1}{\sqrt{\pi \cdot \sigma \cdot \mu \cdot f}}$$

- e_d : skin depth thickness
- σ : conductivity
- μ : permeability
- f : frequency

Passive components

Inductor

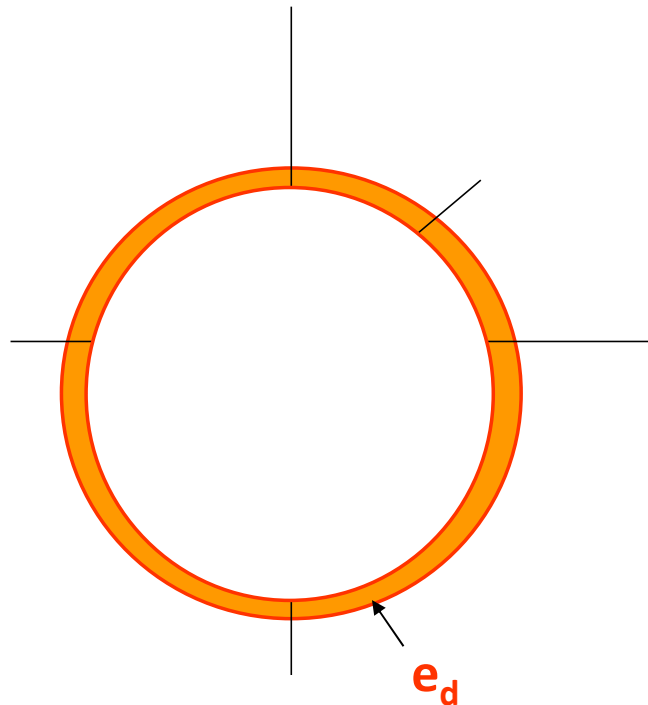
Transformer

Capacitor

Active components

Protections

Skin effect



Copper at 25°C : $\sigma = 6 \cdot 10^7 \text{ S/m}$

$$e_d \approx \frac{66}{\sqrt{f}}$$

➤ 50 Hz : $e_d = 9,4 \text{ mm}$

➤ 100 kHz : $e_d = 0,21 \text{ mm}$

=> *Multi strand wire (Litz wire)*

For AC part of current !

Passive components

Inductor

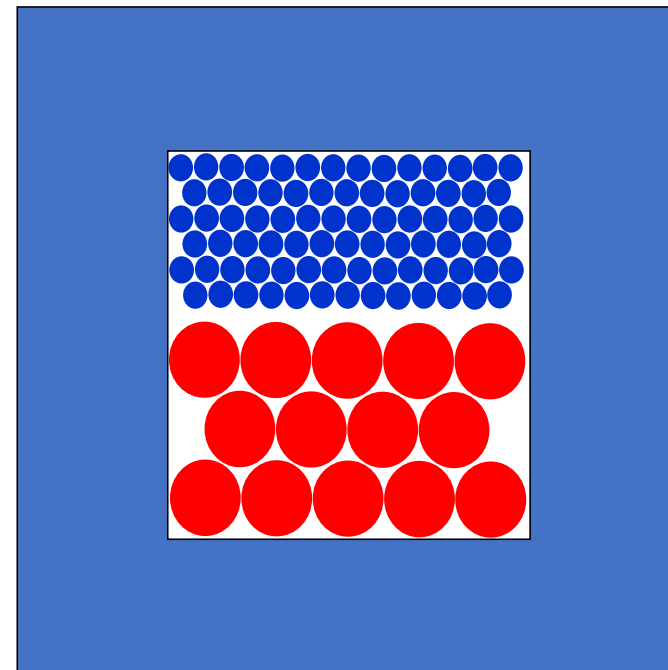
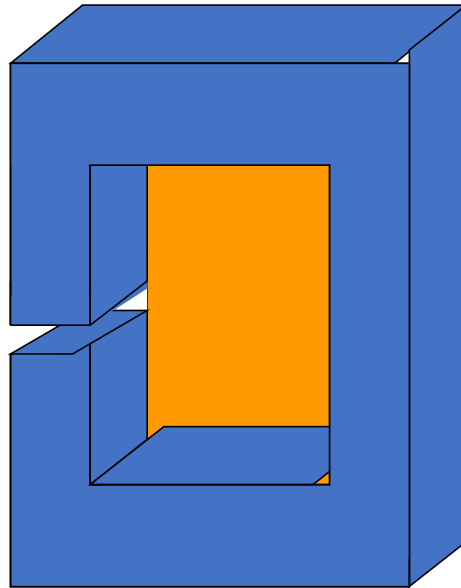
Transformer

Capacitor

Active components

Protections

Winding



$$k_B = 0,7$$

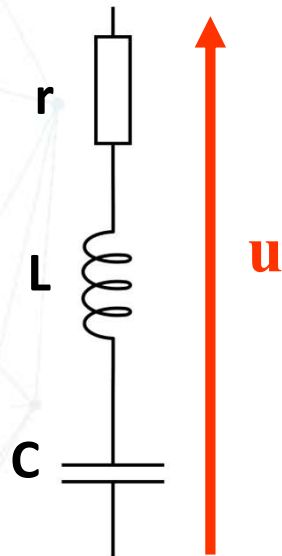
Passive components

Inductor
Transformer
Capacitor

Active components

Protections

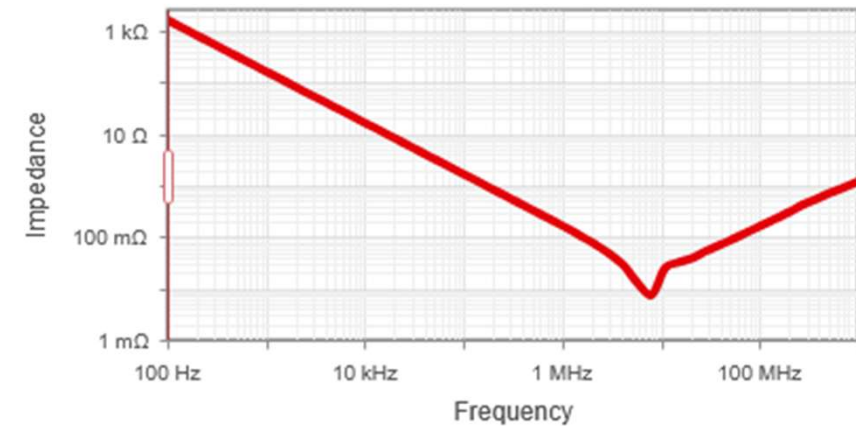
Capacitors



Losses in the capacitor

$$P = r \cdot I_{\text{eff}}^2$$

Rated voltage: U_{eff}



r : ESR (Equivalent Series Resistor)

L : ESL (Equivalent Series Inductor)

Multiple technologies : depends on the frequency

Passive components

Inductor
Transformer
Capacitor

Active components

Protections

Capacitors

Polarized		Unpolarized	
Aluminum	Solid Tantalum	MLCC : Multilayer ceramic capacitors	Film
			
High value		High voltage	
High capacitance		Small size	Self healing
Lifetime depends on temperature	More expensive	Low ESR & ESL	
	Small size	"lower" capacitance	

Capacitor Selection Guidelines for Analog Devices

Fundamentals of Capacitor Technology and their Applications (Würth Elektronik)

Capacitor Selection Guide (Murata)

Passive components

Inductor
Transformer
Capacitor

Active components

Protections

Capacitors

MLCC :

- Class I ceramic capacitor (NP0, C0G ...)
 - Made from ceramic material **not sensitive to temperature** (TiO₂ ...)
 - Low permittivity -> **low capacitance value**
 - Next to no aging
- Class II ceramic capacitor (X5R, X7R, Y5V ...)
 - Made from ceramic materials that are derived from a barium titanate (BaTiO₃) base : **sensitive to temperature**
 - High permittivity : **high capacitance value**
 - High voltage dependency

Passive components

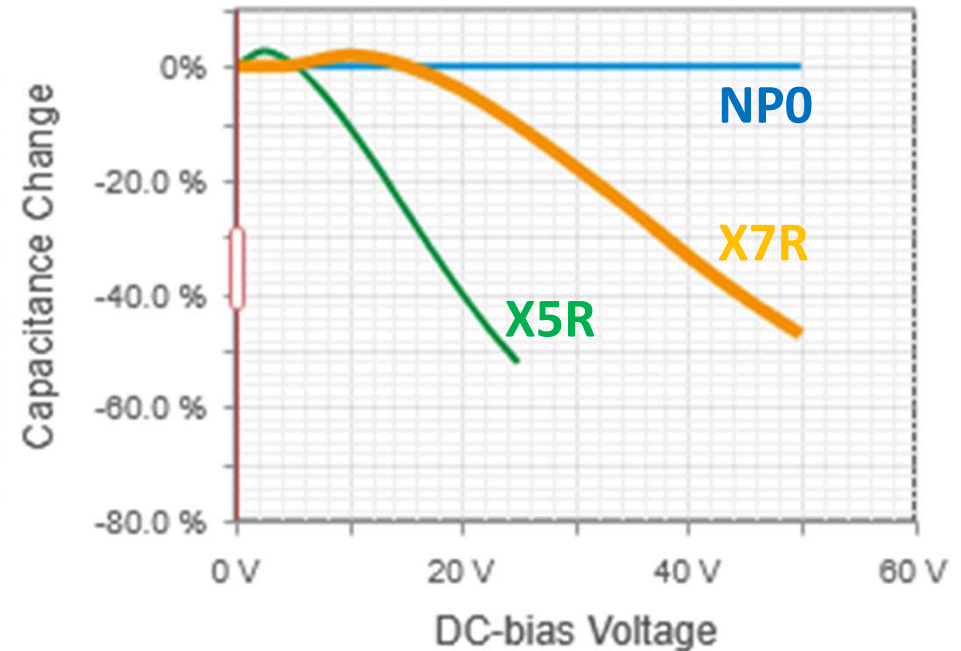
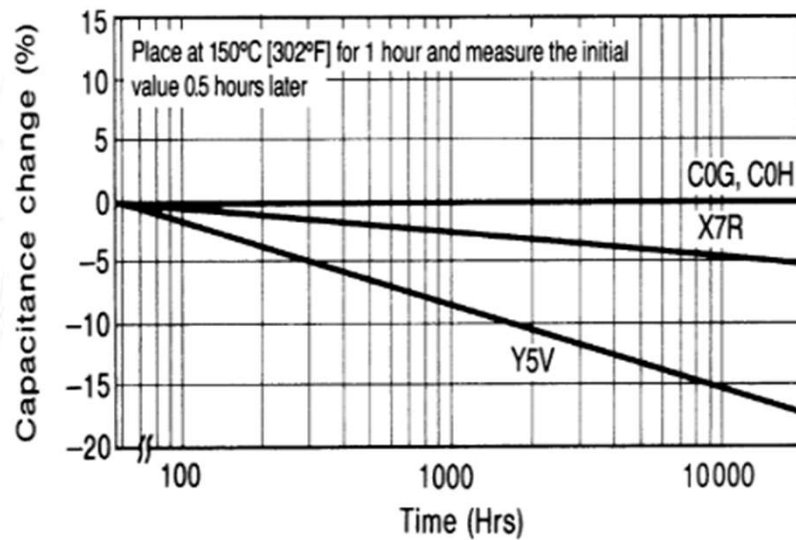
Inductor
Transformer
Capacitor

Active components

Protections

Capacitors

MLCC :



MLCC Application Notes (TDK)

Fundamentals of Capacitor Technology and their Applications (Würth Elektronik)

Alexis MARTIN 30

Passive components

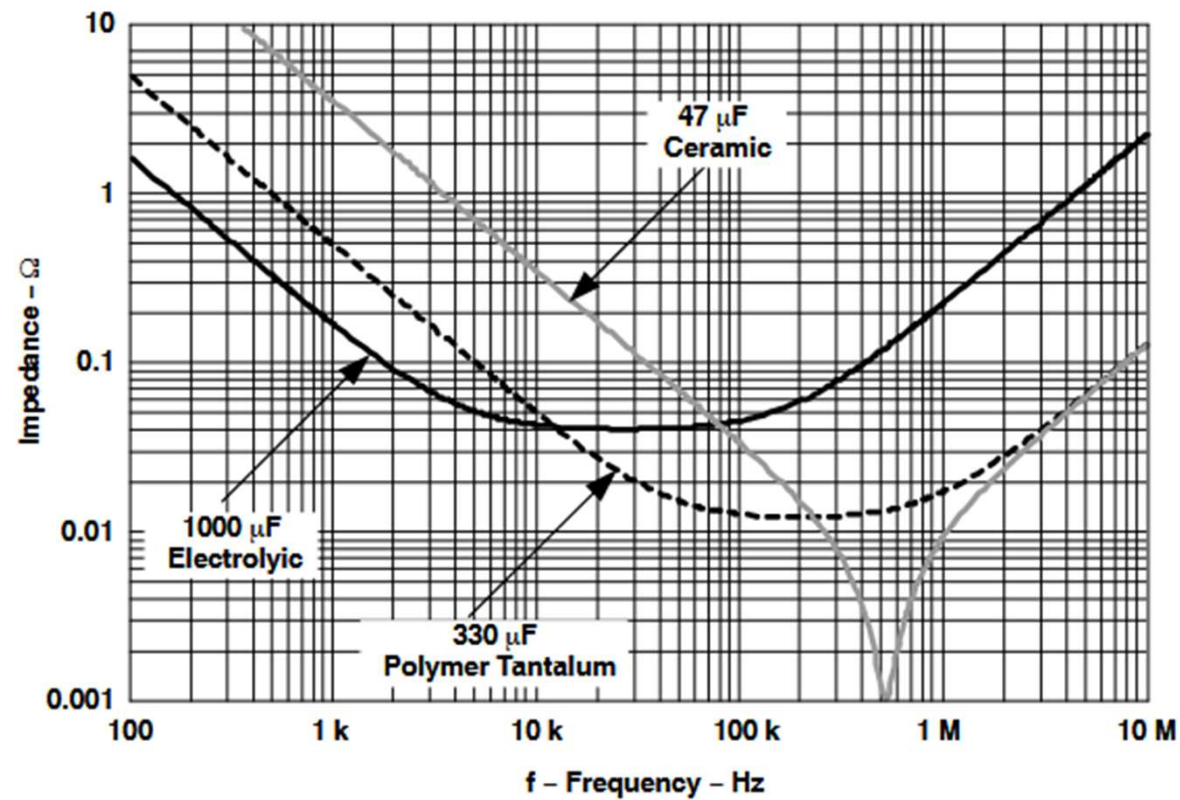
Inductor
Transformer
Capacitor

Active components

Protections

Capacitors

ESR and ESL variations



Input and Output Capacitor Selection (Texas Instrument)

Passive components

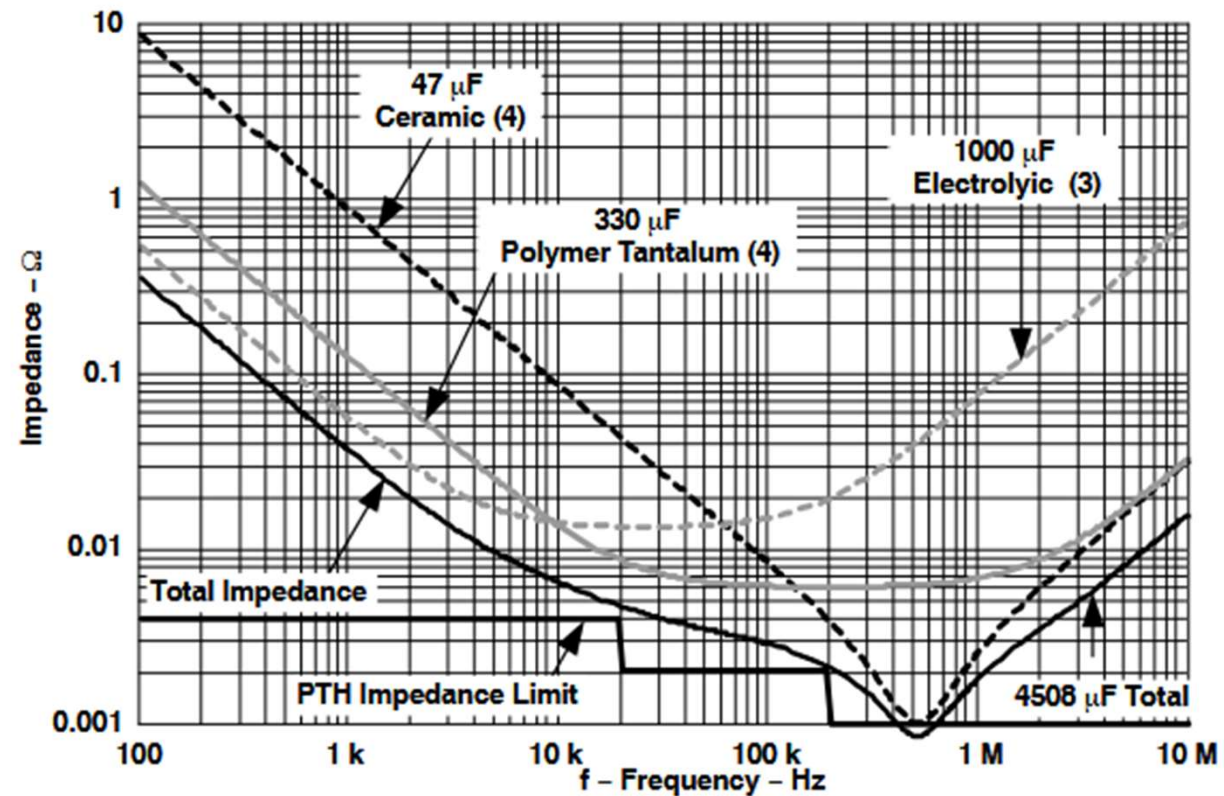
Inductor
Transformer
Capacitor

Active components

Protections

Capacitors

Multiple technologies in parallel



Input and Output Capacitor Selection (Texas Instrument)

Passive
components

Active
components

Protections

Active components

Passive
components

Active
components

Diodes

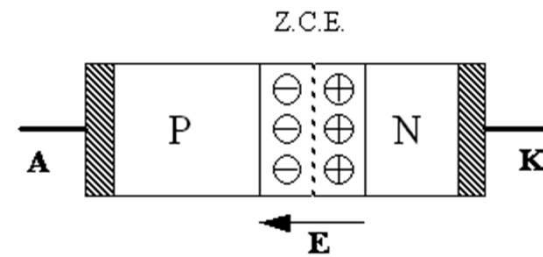
MOS transistor

IGBT transistor

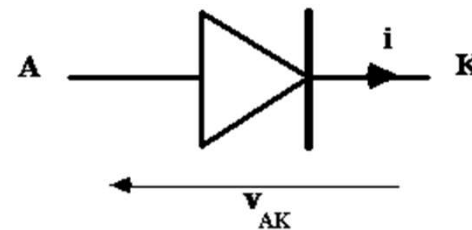
Protections

Active components

PN junction



Diode



$$i = I_S \left(e^{\frac{q \cdot v_{AK}}{k \cdot T}} - 1 \right)$$

Passive
components

Active
components

Diodes

MOS transistor

IGBT transistor

Protections

Diode : static characteristic

I_0 : Average forward current, or I_F , I_{AV}

I_{FM} : Peak forward current

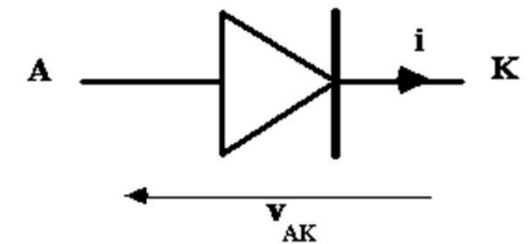
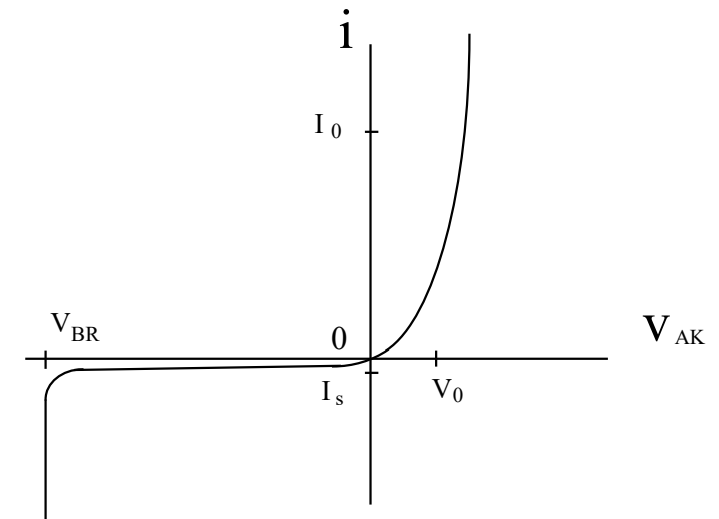
I_{FSM} : Peak forward surge current

I_{RRM} : Reverse repetitive current

V_0 : Forward voltage or V_{FM}

V_{BR} : Breakdown reverse voltage

V_{RRM} : Peak repetitive reverse voltage



Passive
components

Active
components

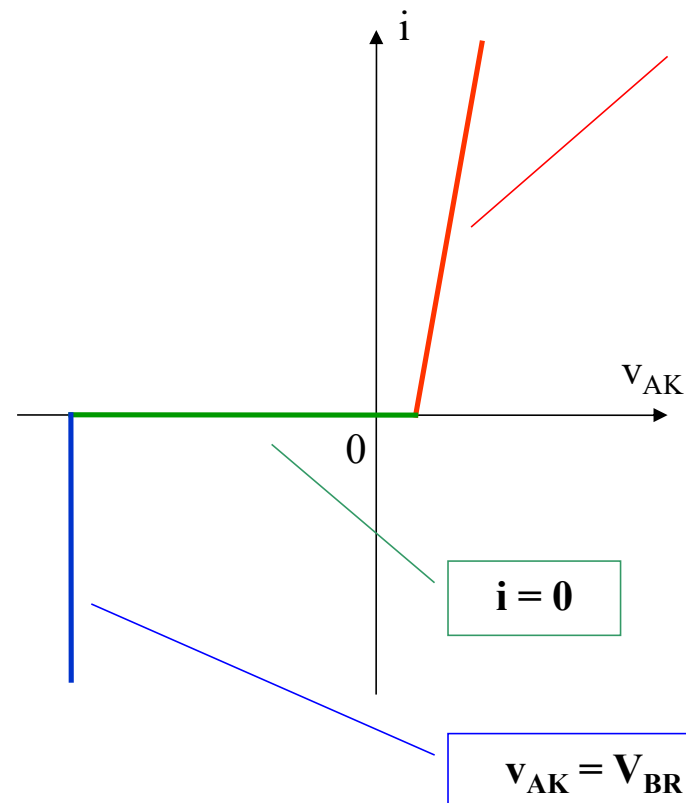
Diodes

MOS transistor

IGBT transistor

Protections

Diode : simplified static characteristics



$$V_{AK} = r_{on} \cdot i + V_0$$

*Linearized characteristic
3 segments*



Conduction loss calculation

$$P_c = r_{on} \cdot I_{eff}^2 + V_0 \cdot I_{av}$$

Passive
components

Active
components

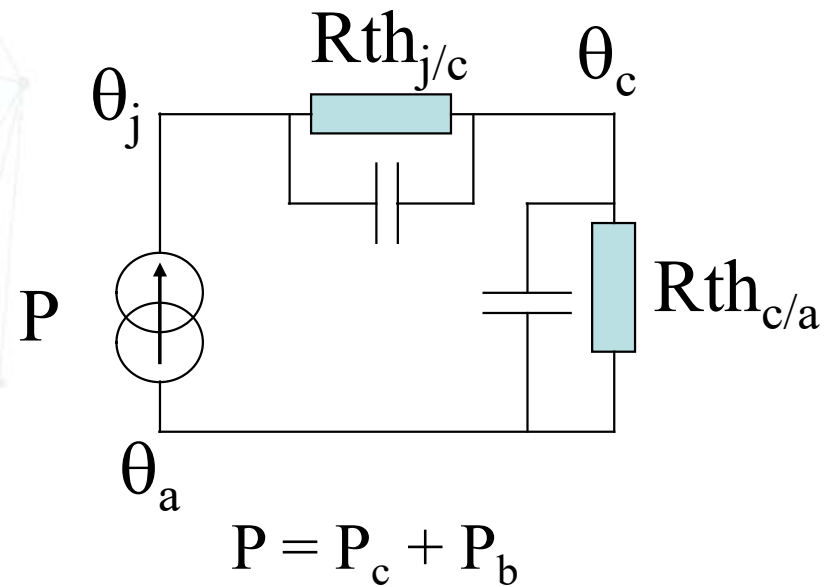
Diodes

MOS transistor

IGBT transistor

Protections

Heat sink



$$\theta_j < \theta_{jmax}$$

$R_{th_{c/a}}$ calculation



Passive
components

Active
components

Diodes

MOS transistor

IGBT transistor

Protections

Specific diode

- *Low threshold diode: Schottky diode*
- *High voltage diode*
- *Power Zener diodes (« Transient-voltage-suppression diode »)*
- *Fast diodes (low Q_R)*
- *Light-emitting diodes (high threshold)*

Passive components

Active components

Diodes

MOS transistor

IGBT transistor

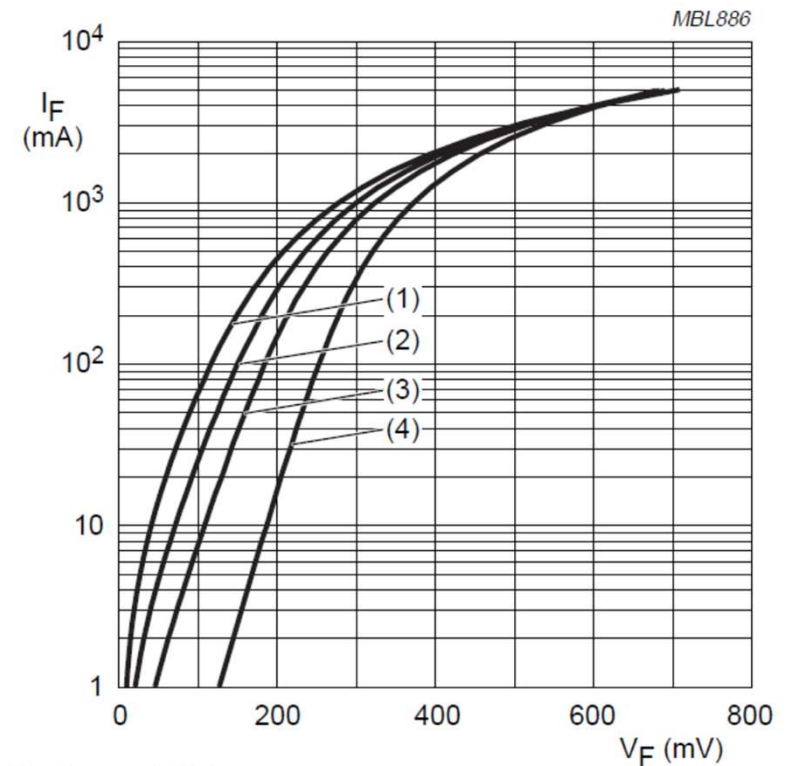
Protections

Schottky diode



Low V_F $\rightarrow$ Low conduction losses

- Used as diodes in converters (instead of PN diodes)
- “low” voltage (max~100V)



- (1) $T_{amb} = 125\text{ }^{\circ}\text{C}.$
- (2) $T_{amb} = 100\text{ }^{\circ}\text{C}.$
- (3) $T_{amb} = 75\text{ }^{\circ}\text{C}.$
- (4) $T_{amb} = 25\text{ }^{\circ}\text{C}.$

Fig.2 Forward current as a function of forward voltage; typical values.

Passive components

Active components

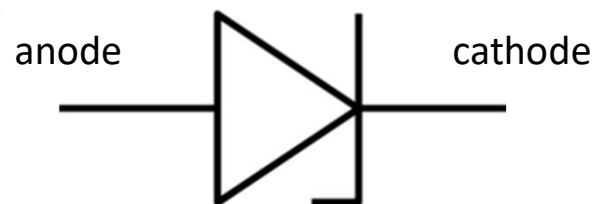
Diodes

MOS transistor

IGBT transistor

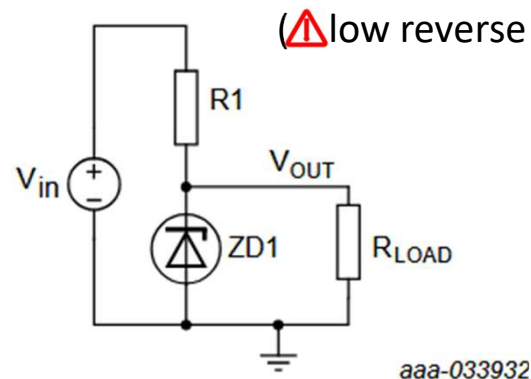
Protections

Zener diode

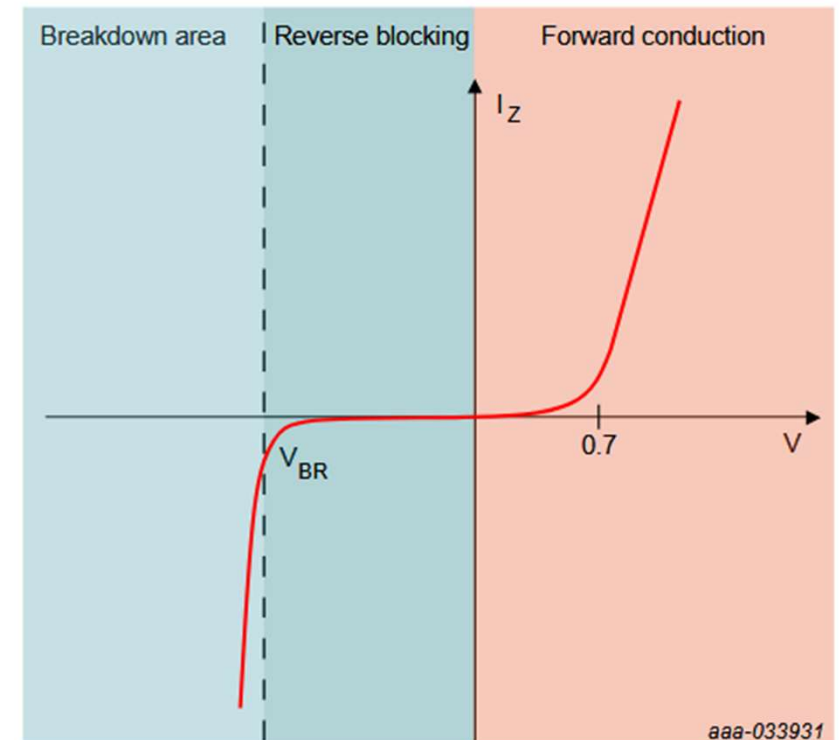


Stable V_{BR}

- Low-cost stabilized voltage
- Signal overvoltage clamping (⚠ low reverse current)



aaa-033932



I-V characteristic of a Zener diode

BZX884S	Sel	Working voltage V_Z (V)		Differential resistance r_{dif} (Ω)			
		$I_Z = 5 \text{ mA}$		$I_Z = 1 \text{ mA}$		$I_Z = 5 \text{ mA}$	
		Min	Max	Typ	Max	Typ	Max
18	B	17.60	18.40	50	225	10	45
	C	16.80	19.10				

AN90031 (Nexperia)

Passive components

Active components

Diodes

MOS transistor
IGBT transistor

Protections

TVS diode

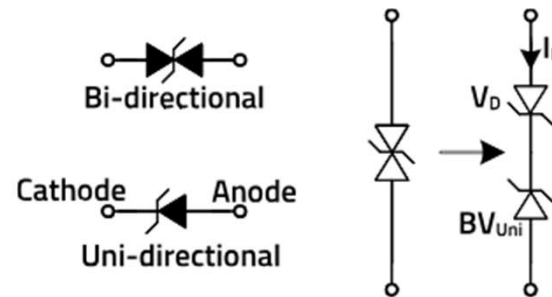
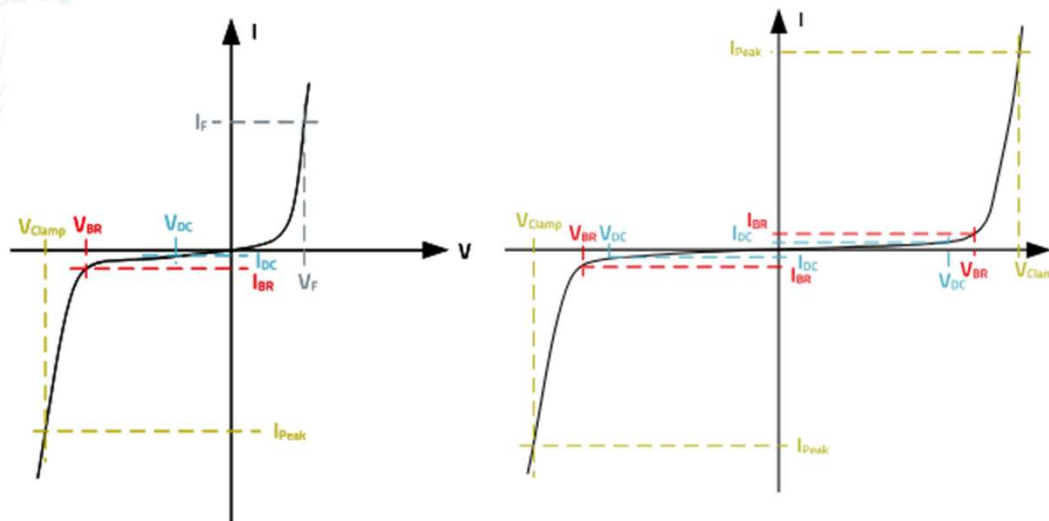


Figure 1: Electrical symbols for the uni- and bidirectional TVS diodes.

Right figure: Functional equivalent of the bidirectional TVS diode.



TVS : Transient Voltage Suppressors

- Protection against high voltage spikes
- ESD protection (USB ...)
- Overvoltage clamping (high peak pulse current)

WE-TVSP Power TVS Diode, 824500101

Properties		Test conditions	Value	Unit	Tol.
DC Operating Voltage	V_{DC}		10	V	max.
(Reverse) Breakdown Voltage	V_{BR}	1 mA	11.7	V	$\pm 5\%$
Clamping Voltage	V_{CLAMP}	I_{DCLAMP}	17	V	max.
(Reverse) Peak Pulse Current	I_{RPM}	10/1000 μs	23.5	A	max.
(Forward) Peak Pulse Current	I_{FPM}		40	A	max.
Leakage Current	I_{LEAK}	VDC	5	μA	max.
Steady State Power Dissipation	P_{DSS}	$T_A = 50^\circ C$	3.3	W	max.
Power Dissipation	P_{DPS}	10/1000 μs	400	W	max.
Polarity	Unidirectional				

ANP143 (Würth Elektronik)

Passive
components

Active
components
Diodes

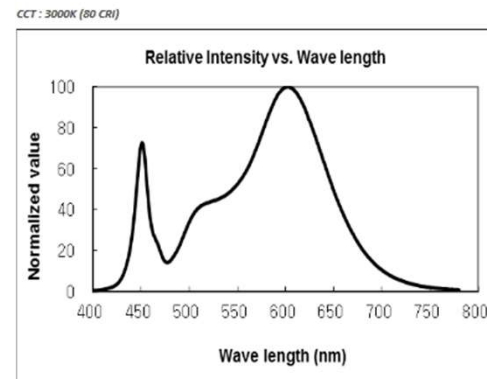
MOS transistor
IGBT transistor

Protections

LED diode



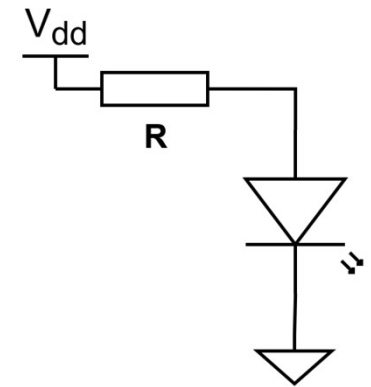
LM301B (Samsung)



« White » LED

b) Electro-optical Characteristics ($I_f = 65 \text{ mA}$, $T_s = 25^\circ\text{C}$)

Item	Unit	Rank	Bin	Min.	Typ.	Max.
Forward Voltage (V_f)	V	XA	AY	2.6	-	2.7
			AZ	2.7	-	2.8
			A1	2.8	-	2.9
Reverse Voltage (@ 5 mA)	V			0.7	-	1.2
Color Rendering Index (R_a)	-			80	-	-
Thermal Resistance (junction to solder point)	$^\circ\text{C}/\text{W}$			-	7.5	-
Beam Angle	$^\circ$			-	120	-



Passive
components

Active
components

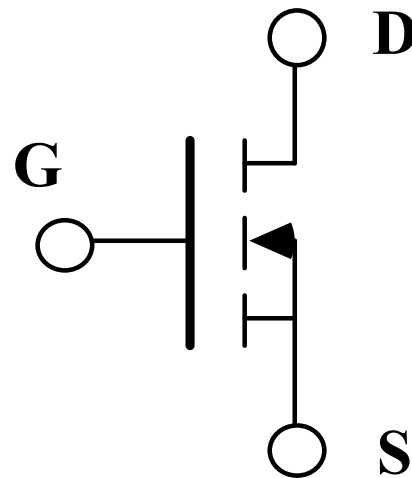
Diodes

MOS transistor

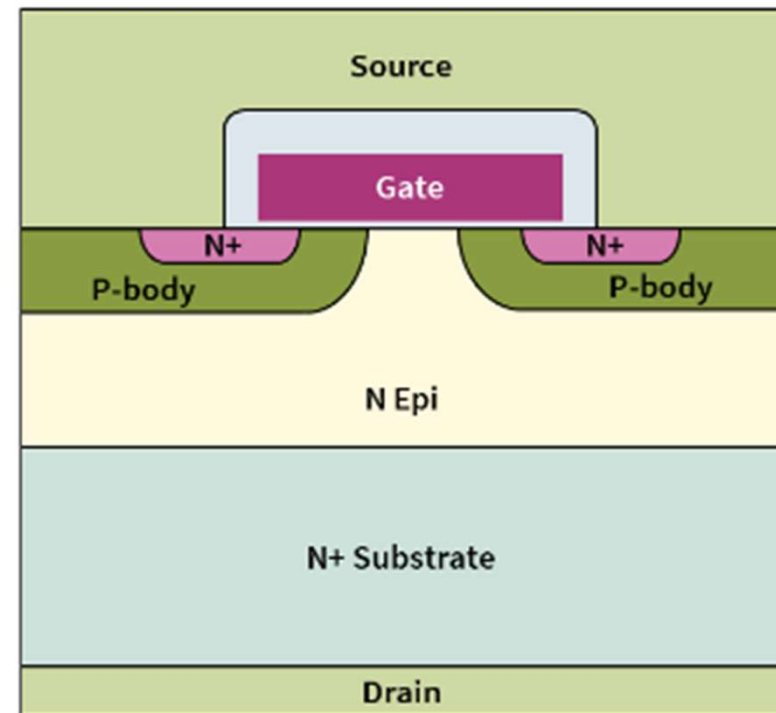
IGBT transistor

Protections

Power MOSFET



Planar MOSFET



Designing with power MOSFETs (Infineon)

Passive
components

Active
components

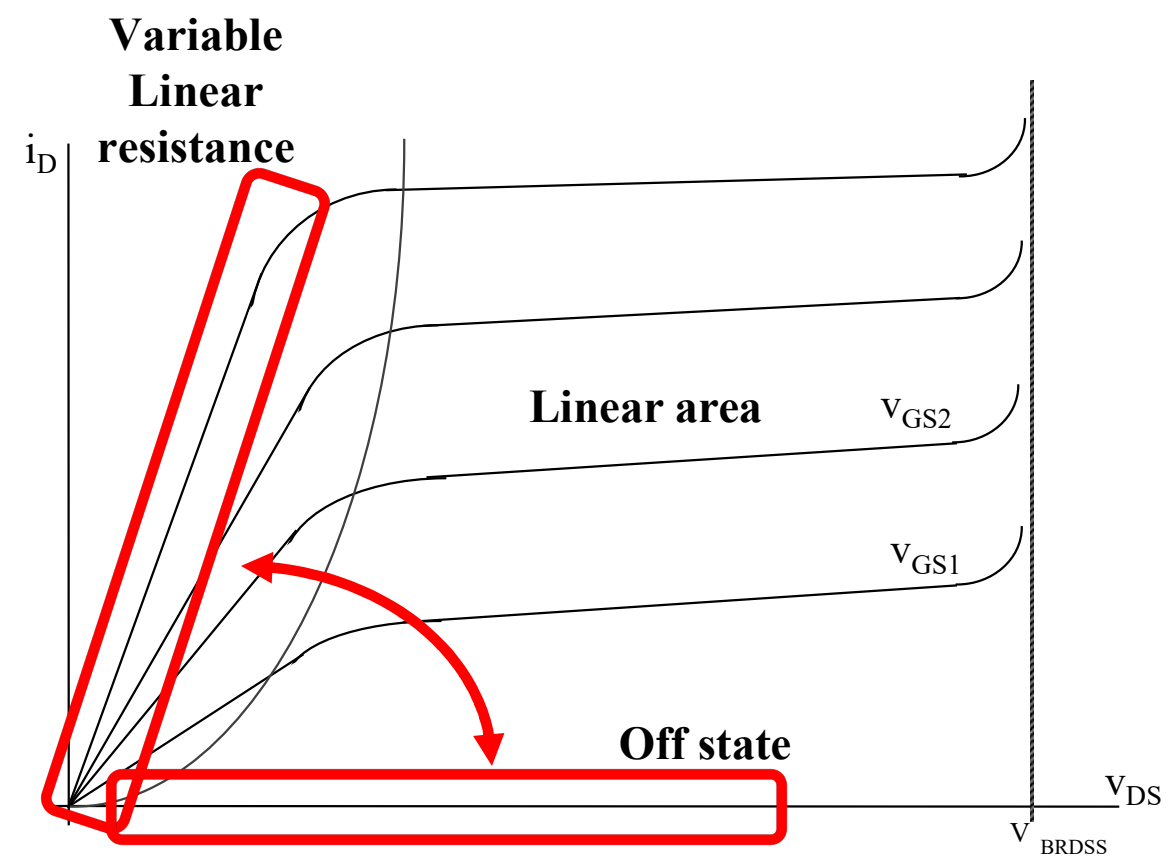
Diodes

MOS transistor

IGBT transistor

Protections

MOSFET characteristics



Passive
components

Active
components

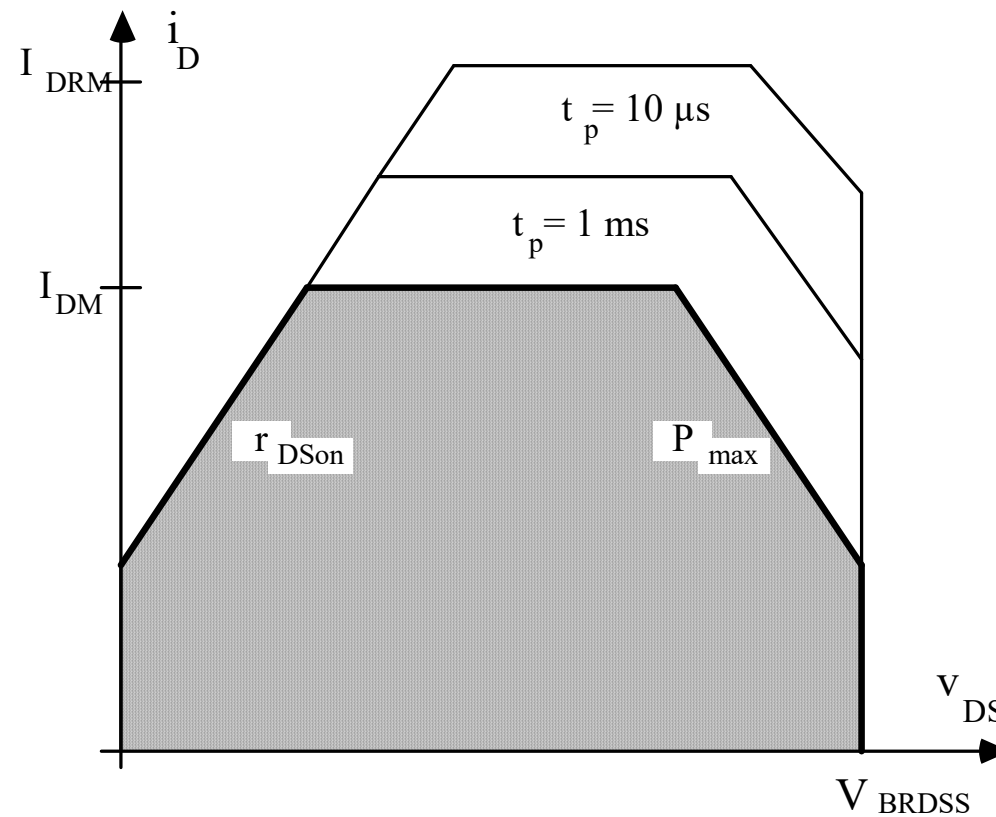
Diodes

MOS transistor

IGBT transistor

Protections

MOS safety area



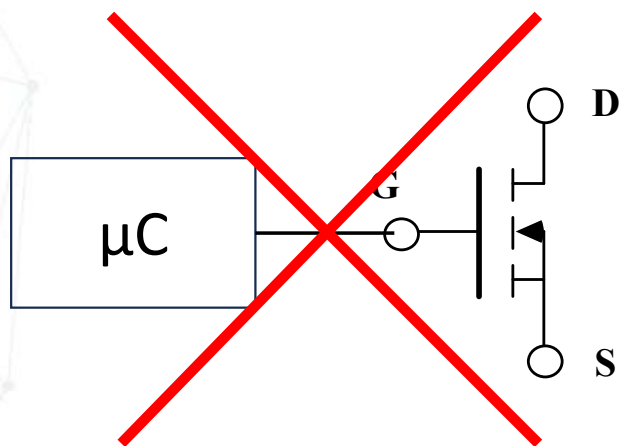
Passive
components

Active
components

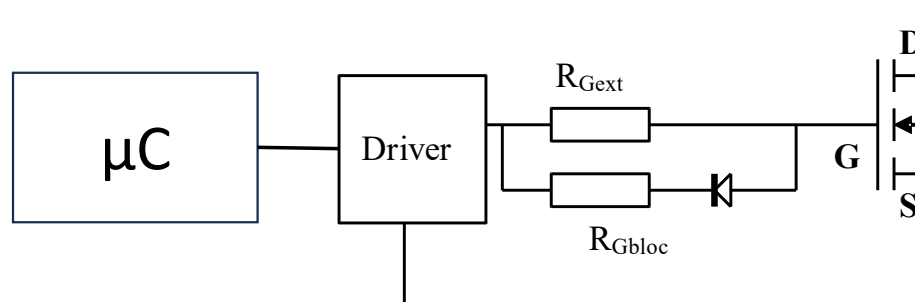
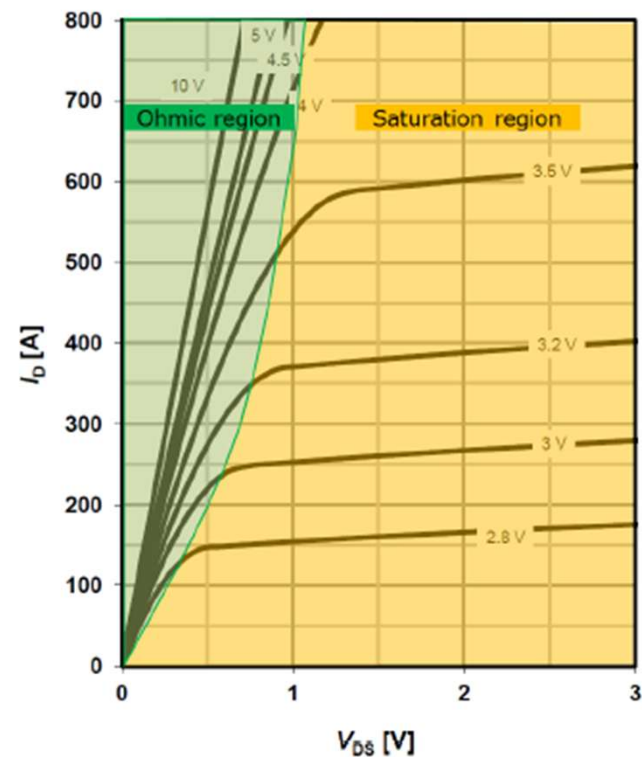
Diodes
MOS transistor
IGBT transistor

Protections

Control specificity



Low V_{gs} => high losses



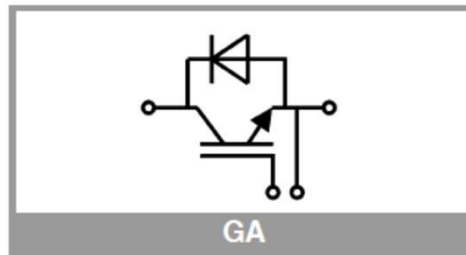
Passive
components

Active
components

Diodes
MOS transistor
IGBT transistor

Protections

Power IGBT



Absolute Maximum Ratings				
Symbol	Conditions		Values	Unit
IGBT				
V _{CES}	T _j = 25 °C		1200	V
I _C	T _j = 175 °C	T _c = 25 °C	1305	A
		T _c = 80 °C	1003	A
I _{Cnom}			900	A

Passive
components

Active
components

Diodes
MOS transistor
IGBT transistor

Protections

Modules

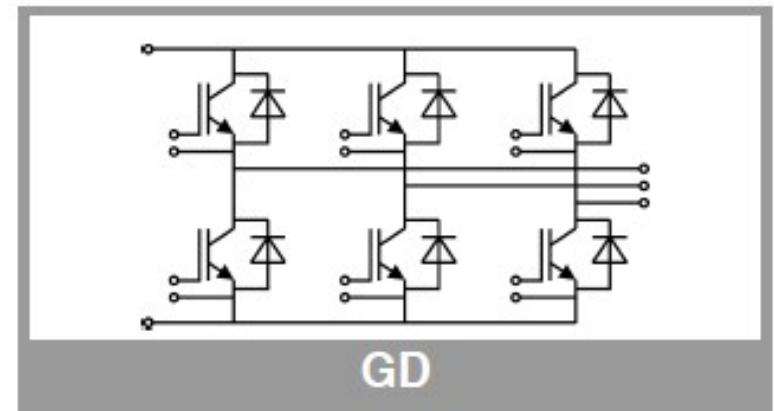


SEMTRANS® 6



SEMiX® 13

1200 V – 39 A at 25°C – $t_f < 200\text{ns}$



1200 V – 231 A at 25°C – $t_f < 500\text{ns}$

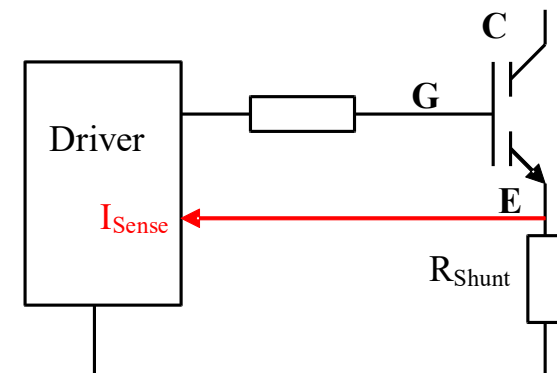
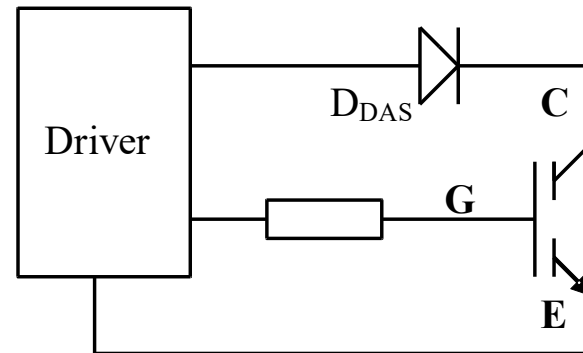
Passive
components

Active
components

Diodes
MOS transistor
IGBT transistor

Protections

Control specificity



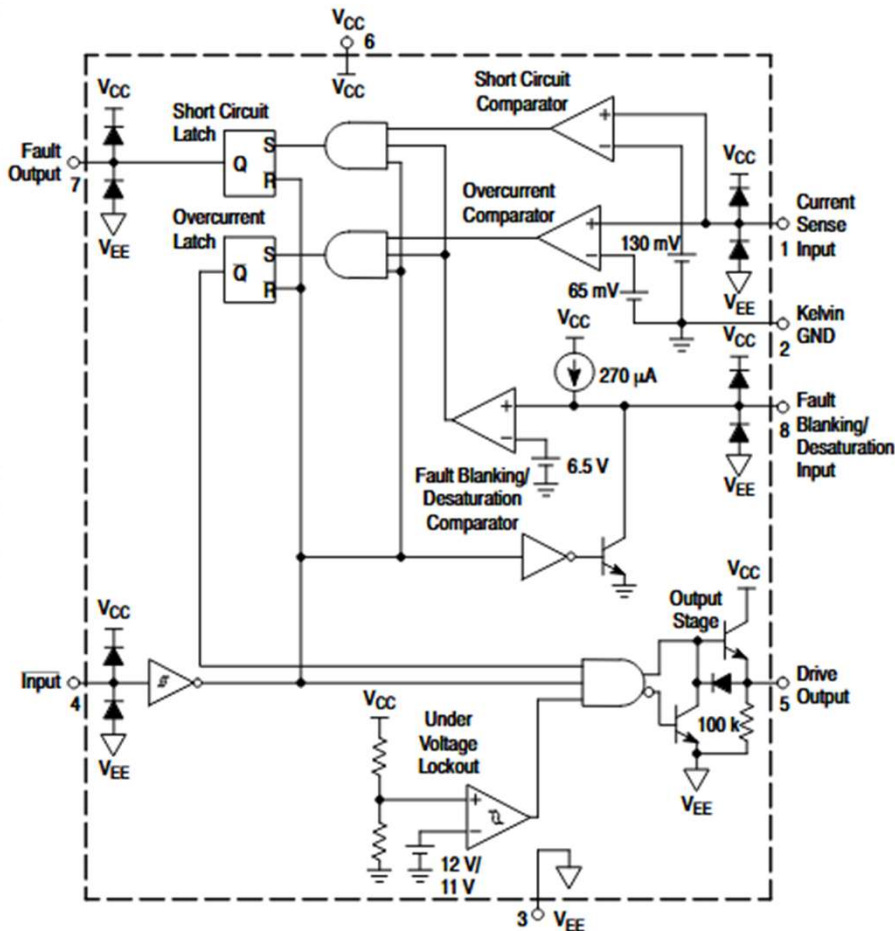
Passive
components

Active
components

Diodes
MOS transistor
IGBT transistor

Protections

IGBT Gate Driver MC33153 (Onsemi)



This device contains 133 active transistors.

Figure 1. Representative Block Diagram

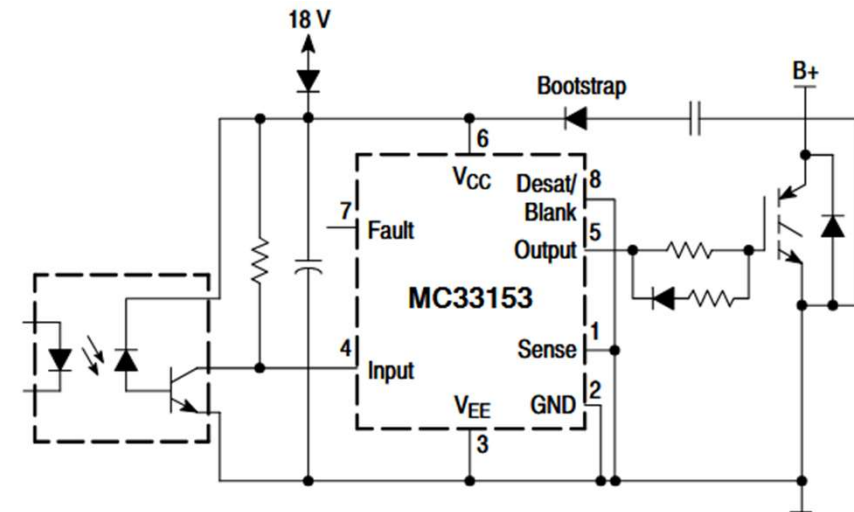


Figure 35. Basic Application



SOIC-8
D SUFFIX
CASE 751



PDIP-8
P SUFFIX
CASE 626

Passive
components

Active
components

Protections

Protections

Passive
components

Active
components

Protections

Fuses

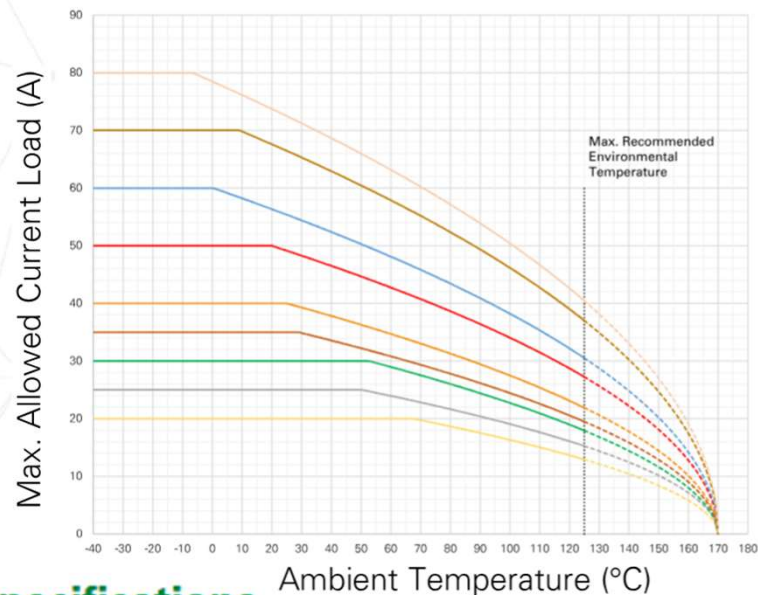
Thermistors

ESD protection

Fuses



- Overcurrent protection

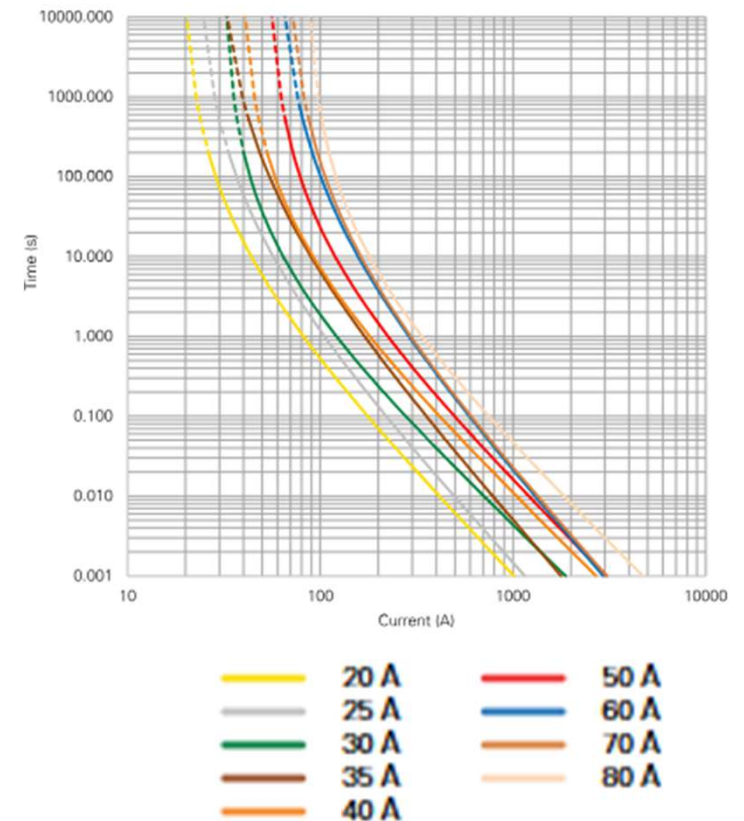


Specifications

Voltage Rating:	32 V DC
Interrupting Rating:	1000 A @ 32 V DC

Part Number	Current Rating (A)	Housing Material Color	Test Cable Size (mm ²)	Typ. Voltage Drop (mV)	Typ. Cold Resistance (mΩ)	Typ. I ² t (A ² s)
0299020._	20	Yellow	4	76	3.1	1100

Time-Current Characteristic



MAXI Series Blade Fuses – Rated 32V (Littlefuse)

Passive
components

Active
components

Protections

Fuses

Thermistors

ESD protection

Fuses



Depending on the reaction time needed, five classes are available on the market (IEC 60127-1 standard):

- FF: very quick acting
- **F: quick acting**
- M: medium time-lag
- **T: time-lag / time delay**
- TT: long time-lag

Fast acting	Time lag / time delay
• No intentional time-delay feature • Fast acting • Hundreds of milliseconds acting • Non inductive loads	• Few seconds delay (motor start ...) • Motor and general protection

Passive
components

Active
components

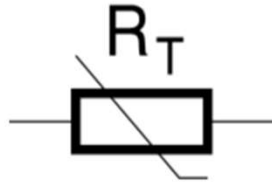
Protections

Fuses

Thermistors

ESD protection

NTC thermistors



NTC : Negative Temperature Coefficient

Commonly referred to as Inrush Current Limiters (ICLs)

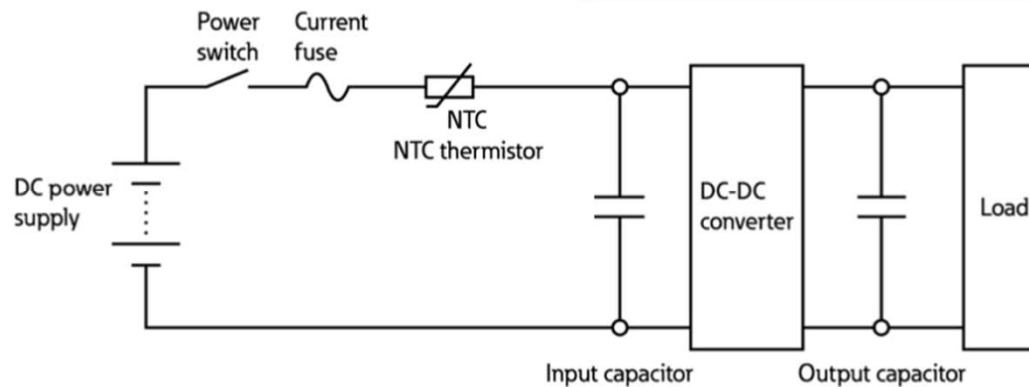
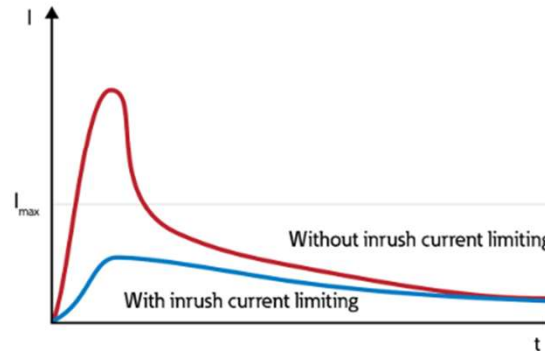
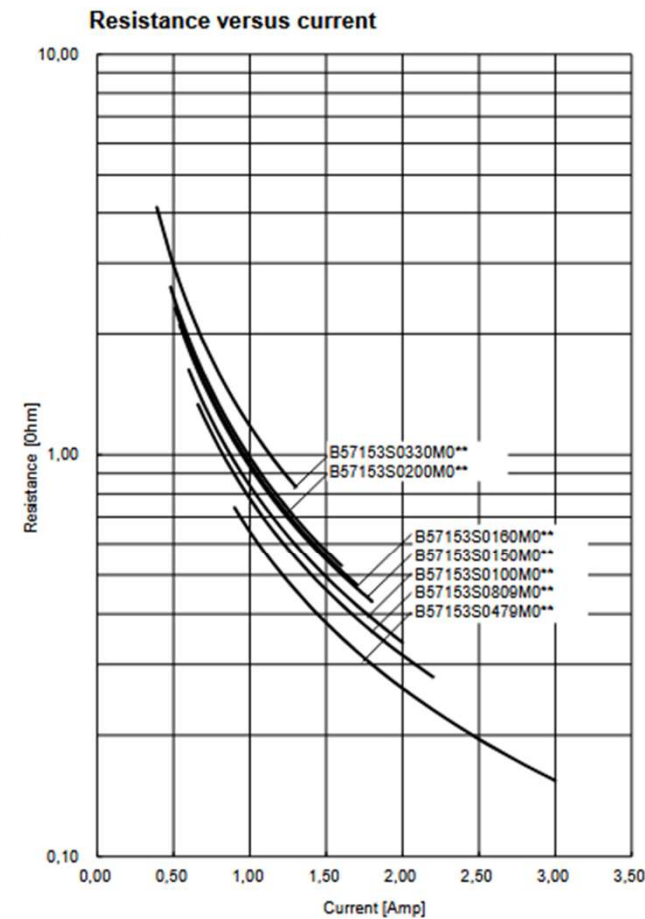


Figure 4: Sample application circuit with power NTC thermistor for a DC-DC converter.

Technical Note ELX1247 (Eaton) + NTC Inrush Current Limiters Application notes (TDK)



Passive
components

Active
components

Protections

Fuses

Thermistors

ESD protection

PTC thermistors

PTC : Positive Temperature Coefficient

- Overcurrent protection
- Temperature sensing

➤ High voltage

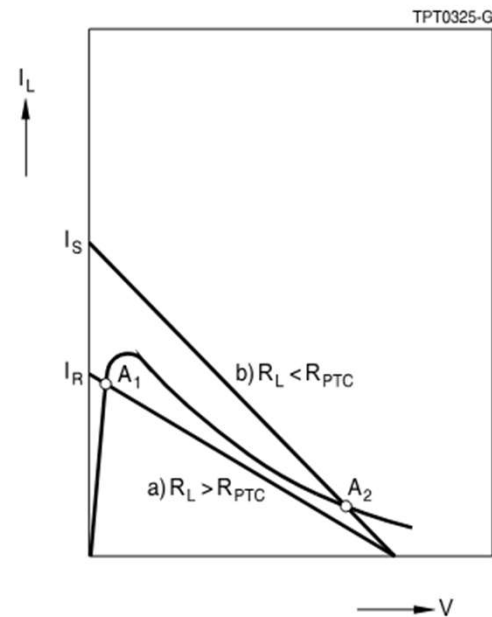
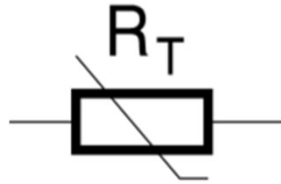


Figure 2

Operating states of a PTC thermistor for overcurrent protection

a) Rated operation

b) Overload operation

R_L	Load resistance
I_S	Switching current
I_R	Rated current
R_{PTC}	PTC resistance

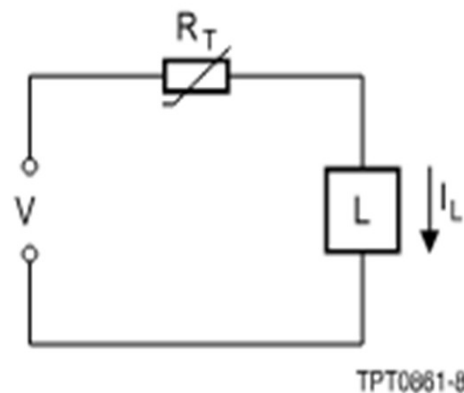


Figure 1

PTC thermistor fuse connected in series with the load

I_L Load current

PTC Thermistors Application notes (TDK)

Passive
components

Active
components

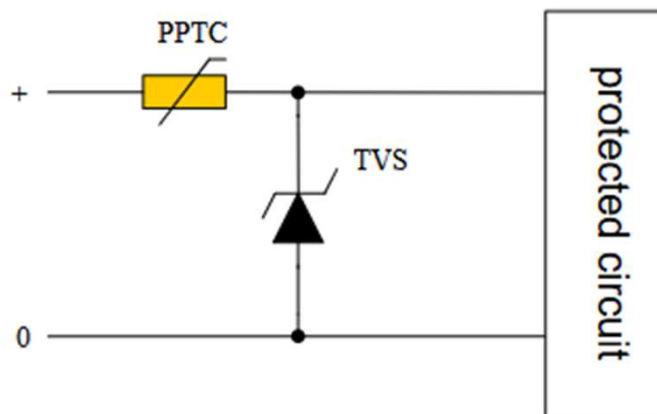
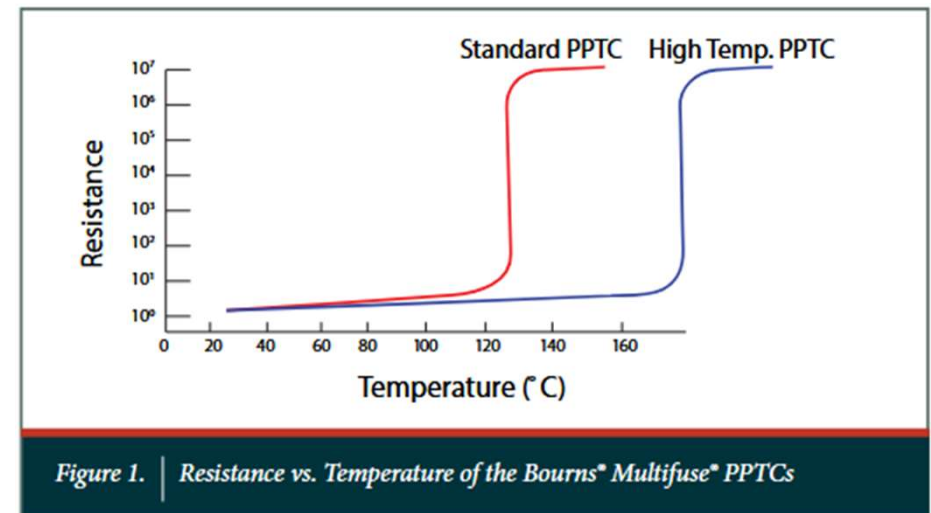
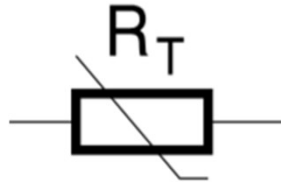
Protections

Fuses
Thermistors
ESD protection

PPTC thermistors

PPTC : Polymeric Positive
Temperature Coefficient

Also known as polyfuse



Part Number	Marking	I_{hold} (A)	I_{trip} (A)
0402L010SL	-	0.10	0.30

Application notes K1467 (Bourns) + POLY-FUSE® and PolySwitch® SMD PPTC Application (Littlefuse)
+ Applications of PPTC in Circuit Protections (Yageo)

Passive
components

Active
components

Protections

Fuses

Thermistors

ESD protection

ESD protection

USB TYPE-C Port

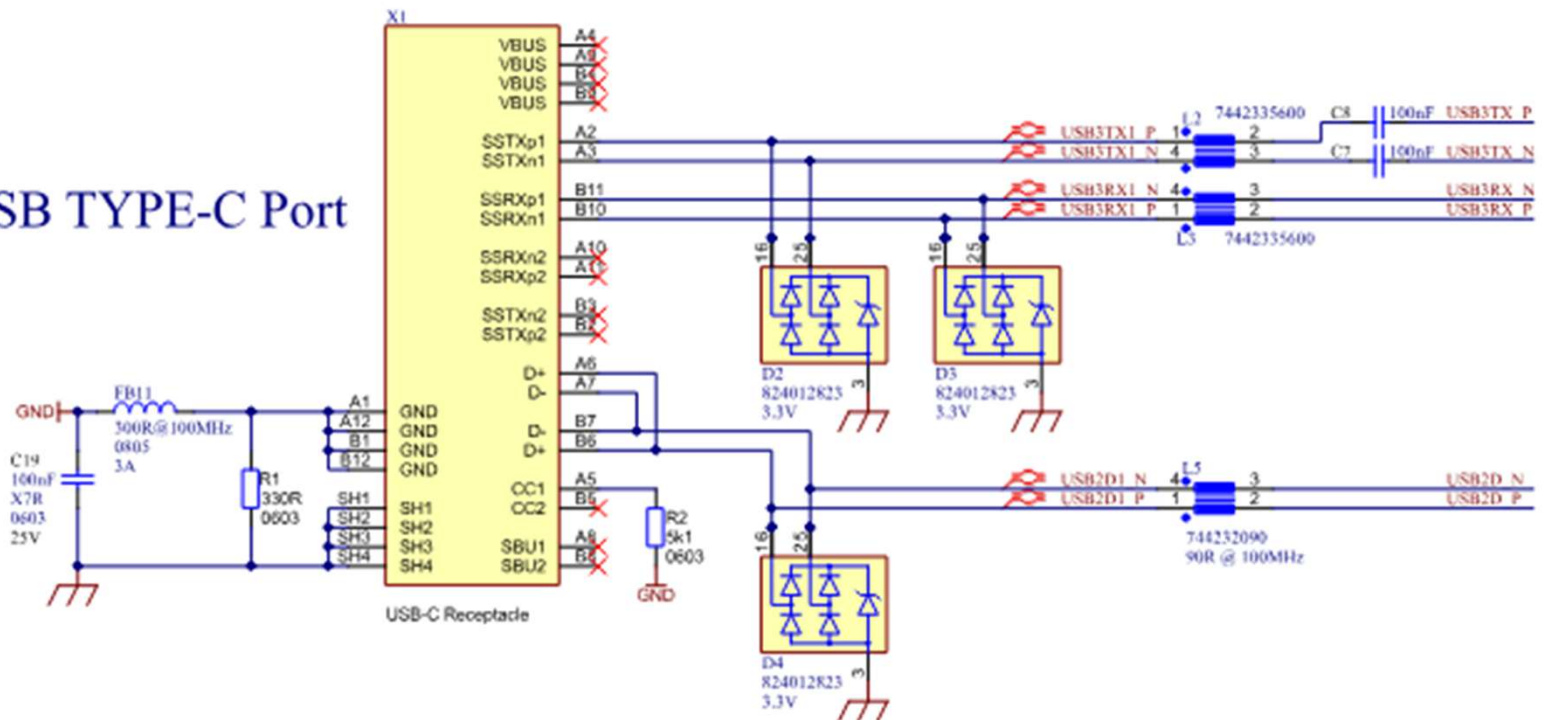


Figure 5: Transient protection at a USB interface with a TVS diode array.

ANP143 (Würth Elektronik)